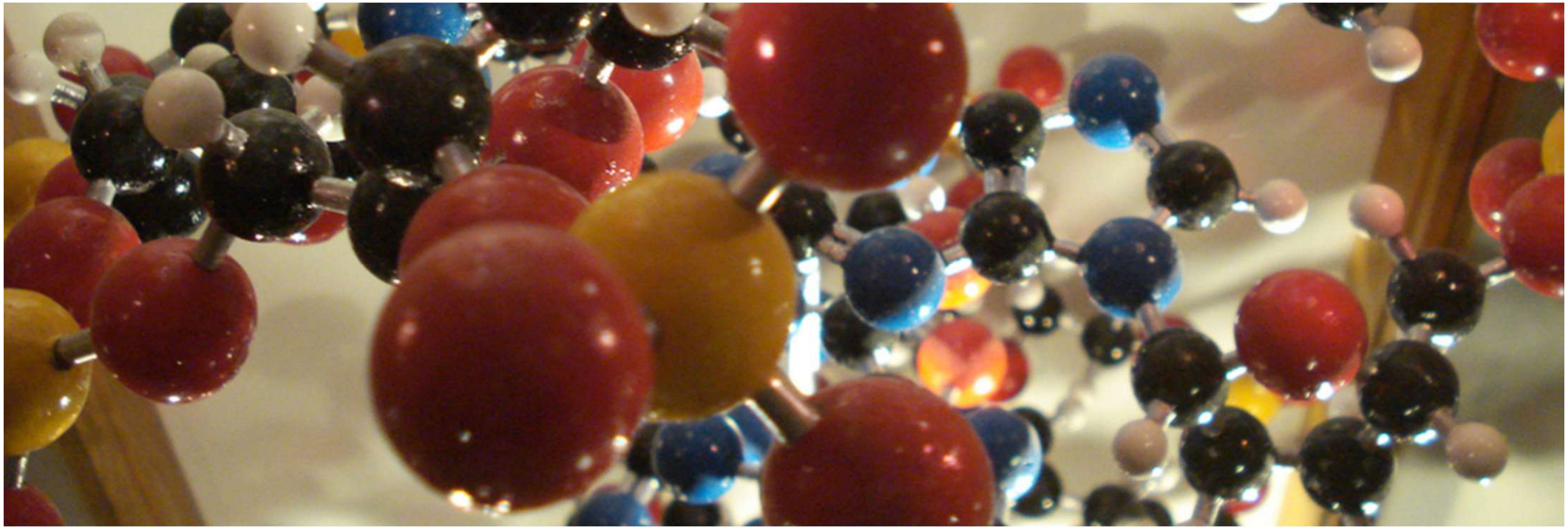
Chapter 2: The Chemistry of Life



Corresponds with OpenStax Biology 2e Chapter 2

Chapter 2: The Chemistry of Life

- **Atoms**: All things are made up of atoms
- **Chemical Bonds**: Atoms link together using chemical bonds to form molecules
- **Water**: Why are molecules of water so important?
- **Hydrocarbons**

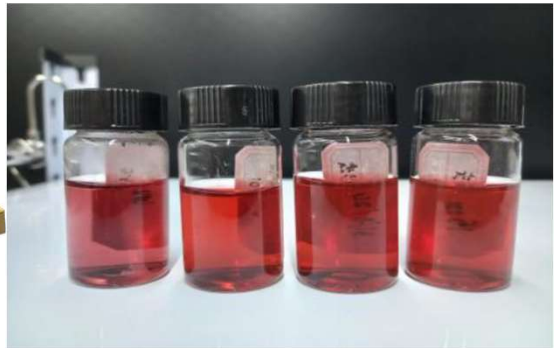
Corresponds with OpenStax Biology 2e Chapter 2

All things are made up of atoms

- All matter that takes up space (solids, liquids, & gases) is made up of **elements**
  - Element = a substance that cannot be broken down chemically into other substances
    - e.g. oxygen, gold, aluminum, calcium, iron, etc.

gold

https://flic.kr/p/THcdWQ



https://pixabay.com/images/id-4131091/

aluminum

https://snl.no/aluminium

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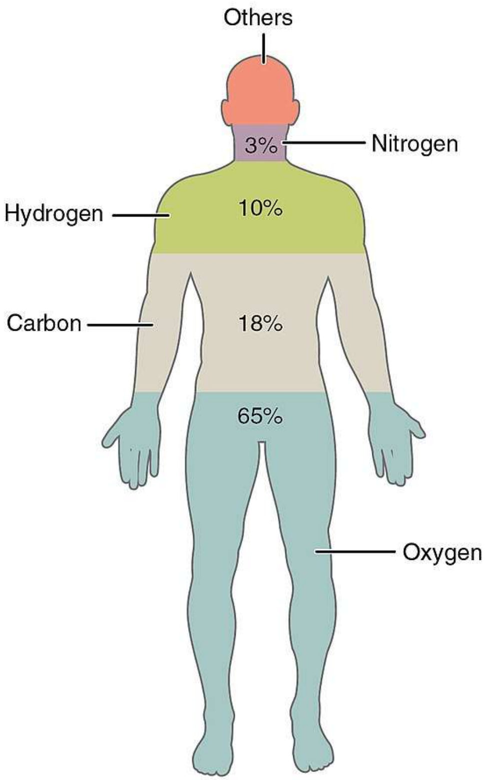
- There are over 100 elements that we know of
- But there are only ~25 elements essential to life
  - if a living thing is deficient in any one element they can become ill or die



Element	Symbol	Percentage in Body
Oxygen	O	65.0
Carbon	C	18.5
Hydrogen	H	9.5
Nitrogen	N	3.2
Calcium	Ca	1.5
Phosphorus	P	1.0
Potassium	K	0.4
Sulfur	S	0.3
Sodium	Na	0.2
Chlorine	Cl	0.2
Magnesium	Mg	0.1
Trace elements include boron (B), chromium (Cr), cobalt (Co), copper (Cu), fluorine (F), iodine (I), iron (Fe), manganese (Mn), molybdenum (Mo), selenium (Se), silicon (Si), tin (Sn), vanadium (V), and zinc (Zn).		less than 1.0

Figure: https://commons.wikimedia.org/wiki/File:201_Elements_of_the_Human_Body-01.jpg

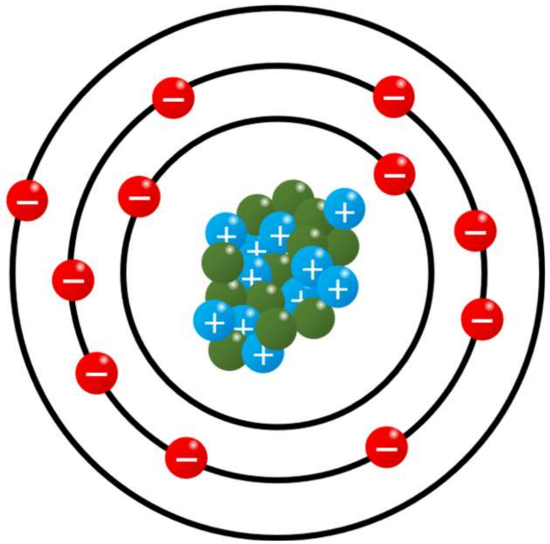
- e.g. an <u>iron</u> deficiency = anemia (low red blood cells), which causes fatigue, impaired immune function, & eventually death

- e.g. a <u>potassium</u> deficiency = muscle cramps, weakness, eventually the heart will stop beating

- e.g. an <u>iodine</u> deficiency = goiters (swollen thyroid glands, located in the neck), depression, & eventually deficits in mental function
  - *You don't need to know these examples, they are just to emphasize the importance of each element, even if you only need small amounts.*

Figure: https://commons.wikimedia.org/wiki/File:Goiter.JPG

- The smallest unit of an element = an **atom**
  – e.g. one unit of oxygen is an atom of oxygen (smallest you can get)



A single atom of the element sodium

"Because they are so long lived, atoms really get around. Every atom you possess has almost certainly passed through several stars and been part of millions of organisms on its way to becoming you. We are each so atomically numerous and so vigorously recycled at death that a significant number of our atoms – up to a billion for each us, it has been suggested – probably once belonged to Shakespeare."

-Bill Bryson in *A Short History of Nearly Everything*

Figure: https://commons.m.wikimedia.org/wiki/File:Sodium_Atom.png

- <u>Atoms are all made up of three components:</u>
  - **Protons** = positive charge (+)
  - **Neutrons** = no charge

  both located inside the *atomic nucleus*

  - **Electrons** = negative charge (-)
    - orbit around the nucleus within *electron shells*
    - *electrons are in constant motion orbiting the nucleus - the shells just represent where they are orbiting*

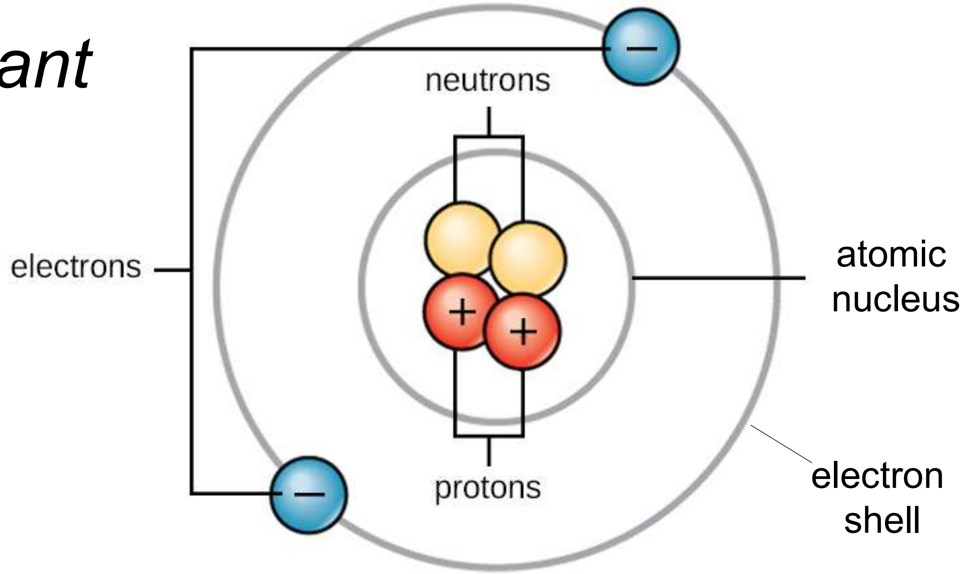
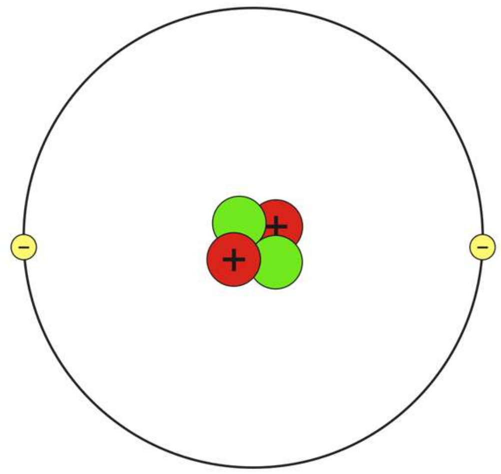


Figure: https://commons.wikimedia.org/wiki/File:OSC_Microbio_00_AA_atom.jpg

- Atoms are electrically <u>neutral</u> because the number of protons (+) is equal to the number of electrons (-)

- Other than hydrogen (a special case), a regular atom has the same number of neutrons
  - e.g. helium atom = 2 protons, 2 neutrons, 2 electrons

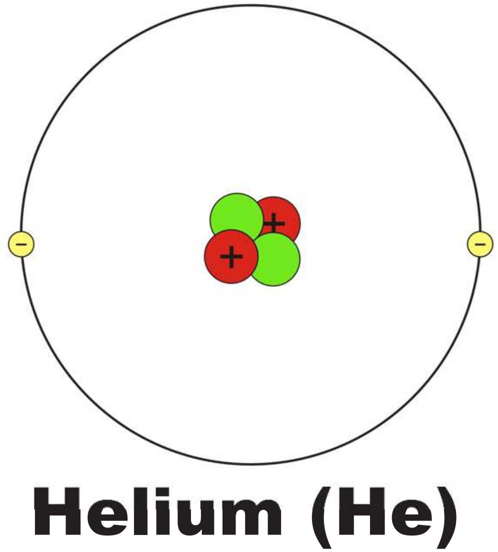
- *If an atom does not have the same number of everything, it will have a special name – we'll talk about these soon*



Helium (He)

Figure: https://commons.wikimedia.org/wiki/File:Helium-Bohr.svg

- **Protons**
  - Each element has a unique number of protons
    - e.g. hydrogen ALWAYS has one
    - e.g. helium ALWAYS has two
    - e.g. carbon ALWAYS has six
  - If you change the number of protons, you change the element completely
    - *e.g. if you add a proton to hydrogen, it becomes helium*



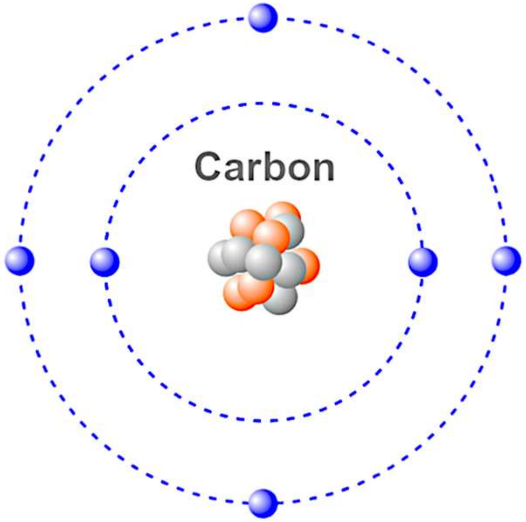
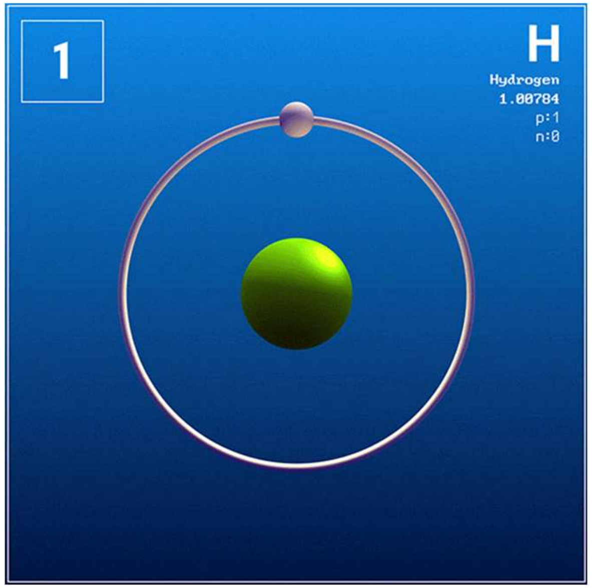


Figure: https://commons.wikimedia.org/wiki/File:Helium-Bohr.svg

Figure: https://pixabay.com/images/id-4426054/

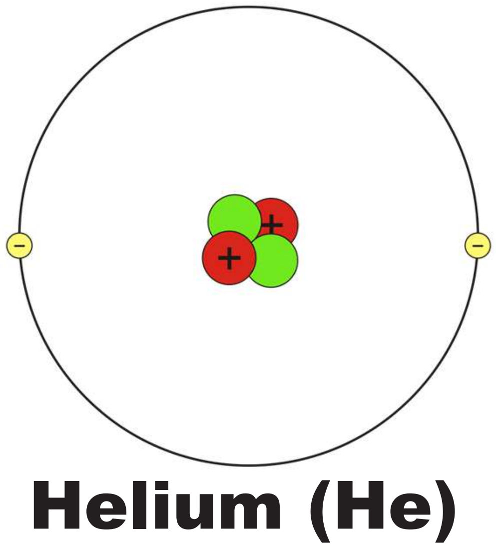
- Because an atom of hydrogen always has one proton no matter what, we can define hydrogen by this unchanging fact – we call this its atomic number
  - **Atomic number** = the number of protons an atom has

Hydrogen has an atomic number of 1



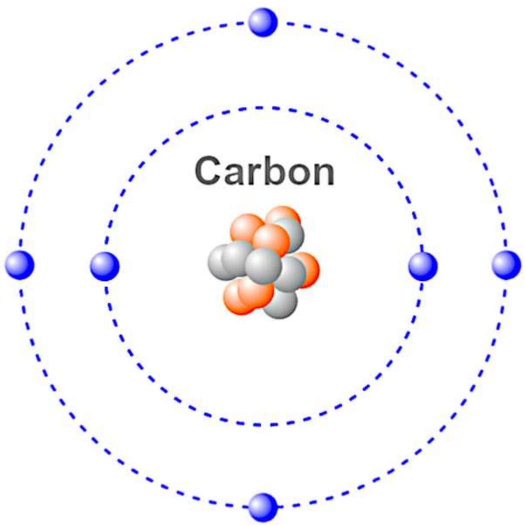
https://commons.wikimedia.org/wiki/File:Hydrogen_GIF.gif

Helium = 2



https://commons.wikimedia.org/wiki/File:Helium-Bohr.svg

Carbon = 6



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- On a periodic table of elements, you'll always see the atomic number for each element listed

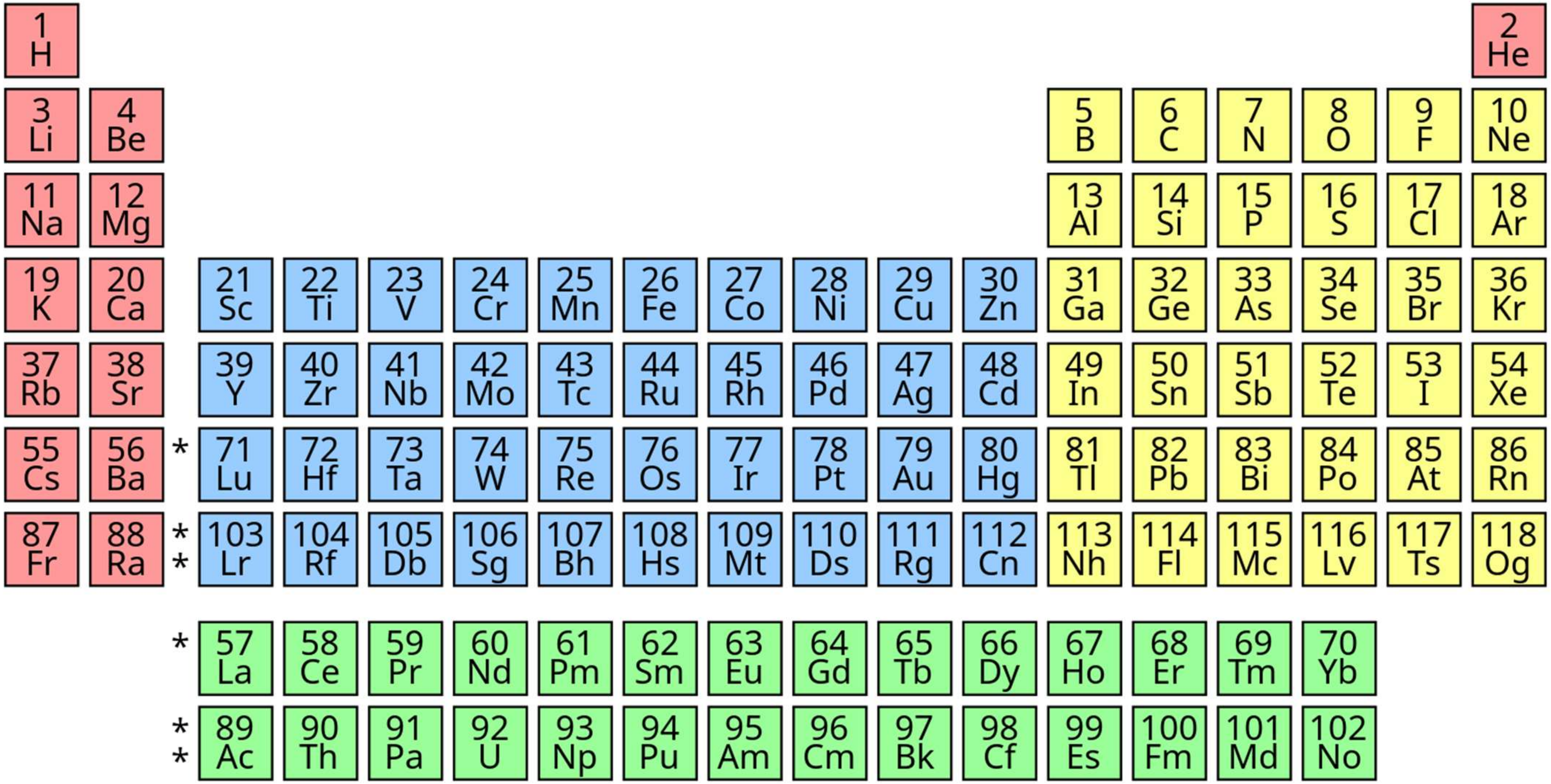
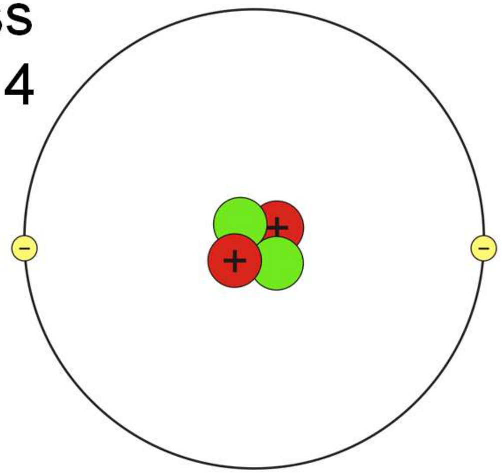


Figure: https://commons.wikimedia.org/wiki/File:Simple_Periodic_Table_Chart-blocks.svg

- We can also describe an atom by its atomic mass
  - **Atomic mass** = number of protons + neutrons
    - *Electrons are so light, we usually ignore them in biology*

Atomic mass of helium = 4



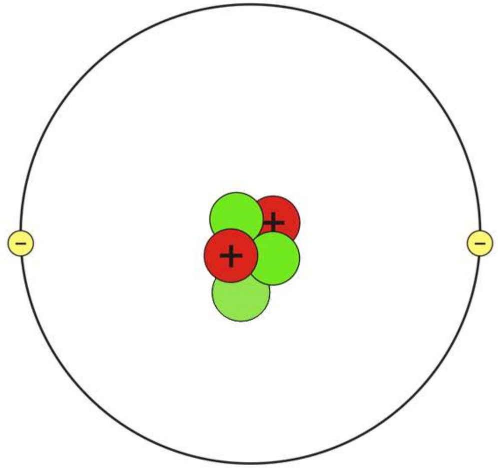
Helium (He)

Protons, Neutrons, and Electrons

	Charge	Mass (amu)	Location
Proton	+1	1	nucleus
Neutron	0	1	nucleus
Electron	−1	0	orbitals

Figure: https://commons.wikimedia.org/wiki/File:Helium-Bohr.svg
Figure from OpenStax Biology 2e

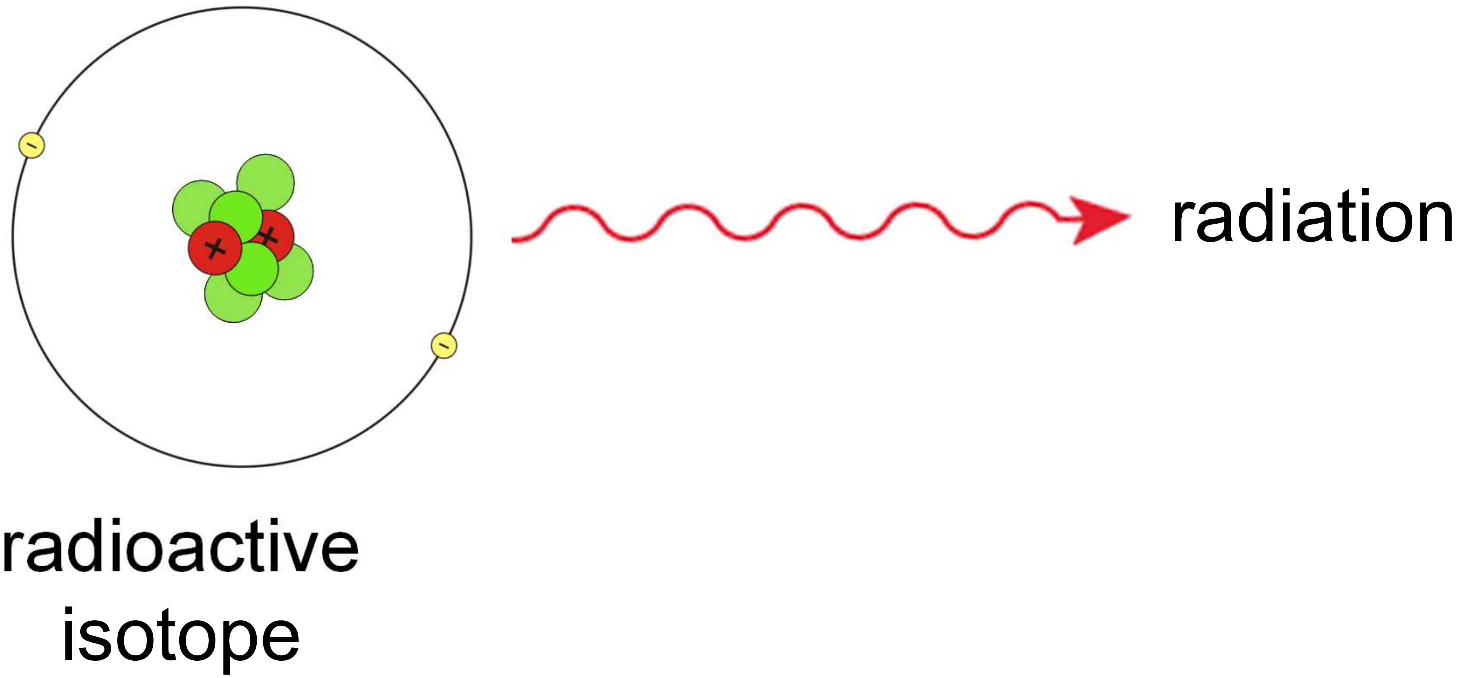
- **Neutrons**
  - The number of protons never changes, but the number of neutrons an atom has can change
    - A regular helium atom has
      2 P+, 2 N, & 2 E-
  - An **isotope** = an atom with a different number of neutrons
    - e.g. helium with
      2 P+, **3 N**, & 2 E-



Helium (He)

Figure: https://commons.wikimedia.org/wiki/File:Helium-Bohr.svg

- Why are isotopes important in biology?
  - Some isotopes are stable, but some are unstable
    - Unstable isotopes are **radioactive**
      - They break apart, releasing radiation



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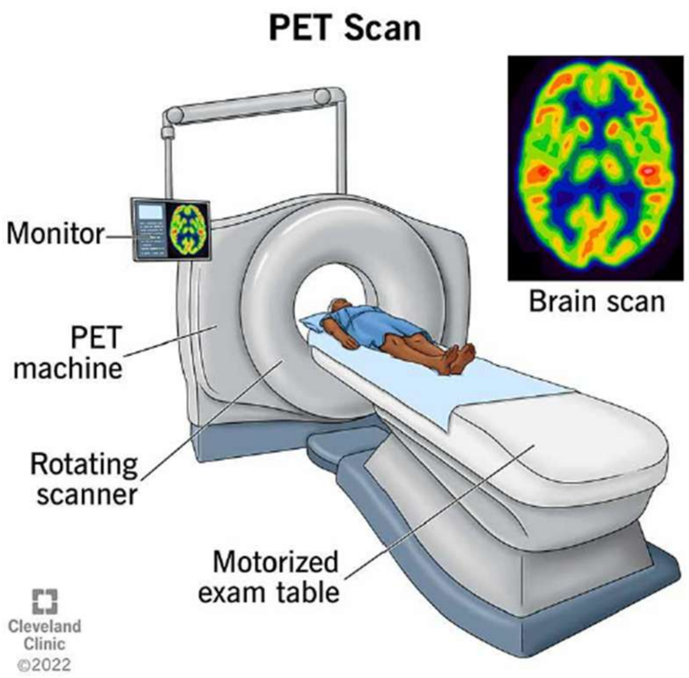
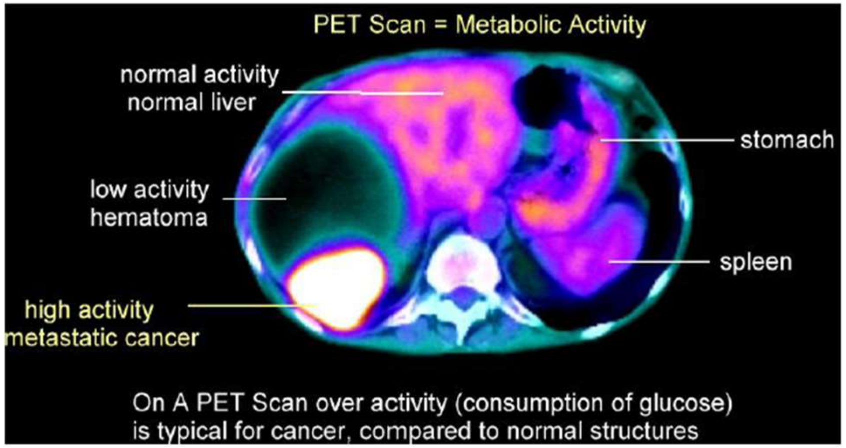


- **<u>How radioactivity / radiation benefits science:</u>**
  - Harnessed for energy
  - Treatment for cancer
  - Determining the age of fossils
  - Medical imaging (PET scans)
    - PET scan = low dose of radioactive glucose injected into patient
      - *Positron Emission Tomography*
      - *glucose is broken down for energy, releasing radio-activity, which we can measure*

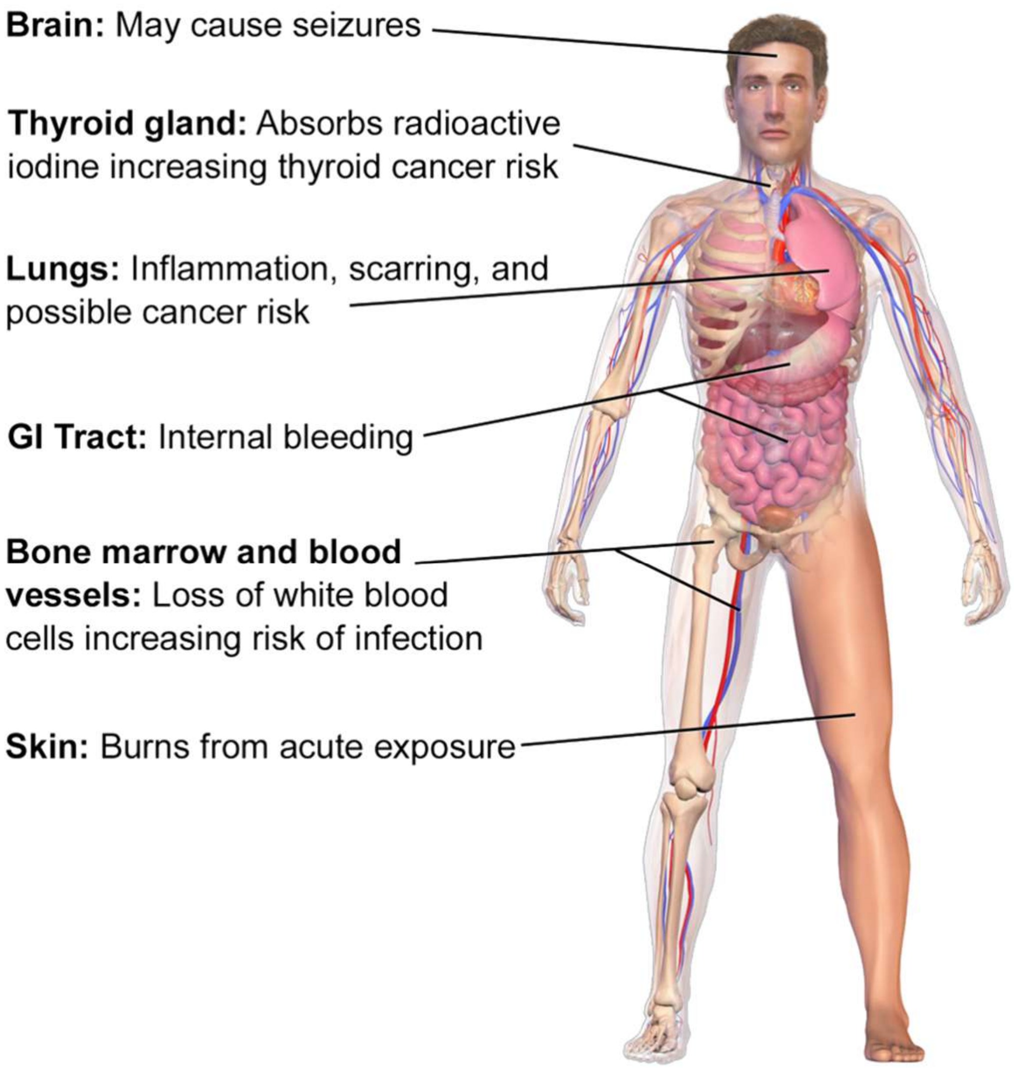


Red indicates high radioactivity, showing a cancerous tumor using a lot of energy

http://www.aboutcancer.com/pet_scan.htm

Figure: https://commons.wikimedia.org/wiki/File:PET-CT_scanning_of_lymph_node_metastases_in_cancer.jpg

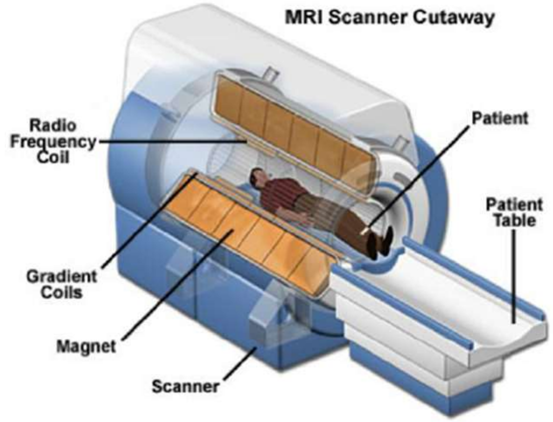
- Too much exposure to radiation can cause:
  - cancer
  - radiation sickness
  - severe birth defects
  - even death

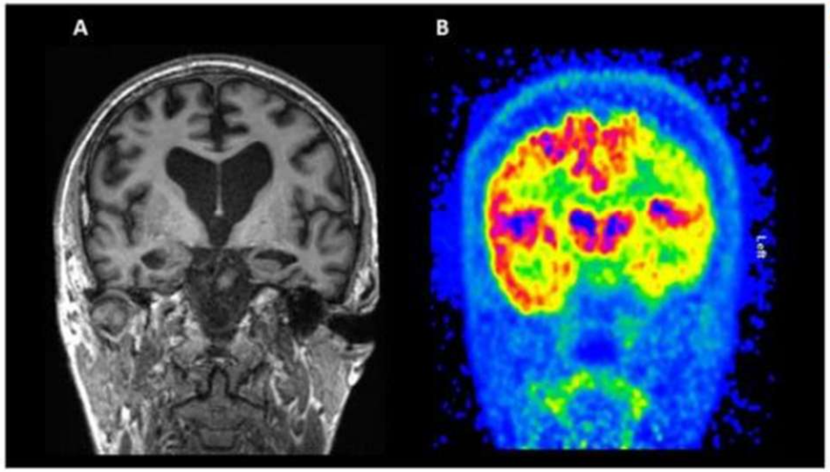


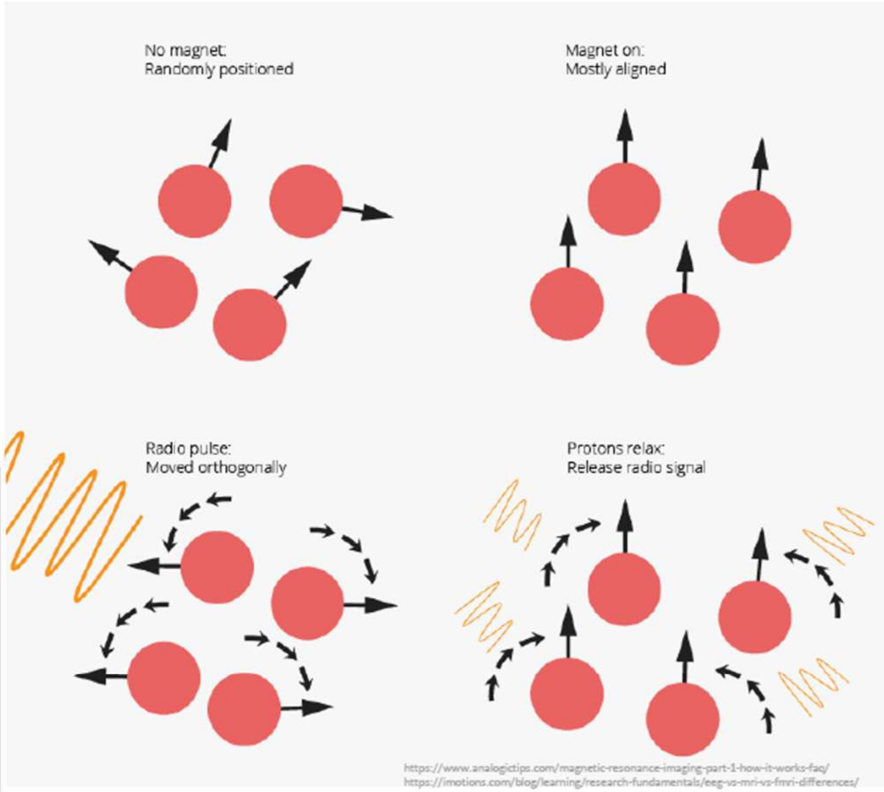
Selected Risks from Radiation Sickness

Figure: https://commons.wikimedia.org/wiki/File:Radiation_Sickness.png

MRI：Nuclear Magnetic Resonance Imaging

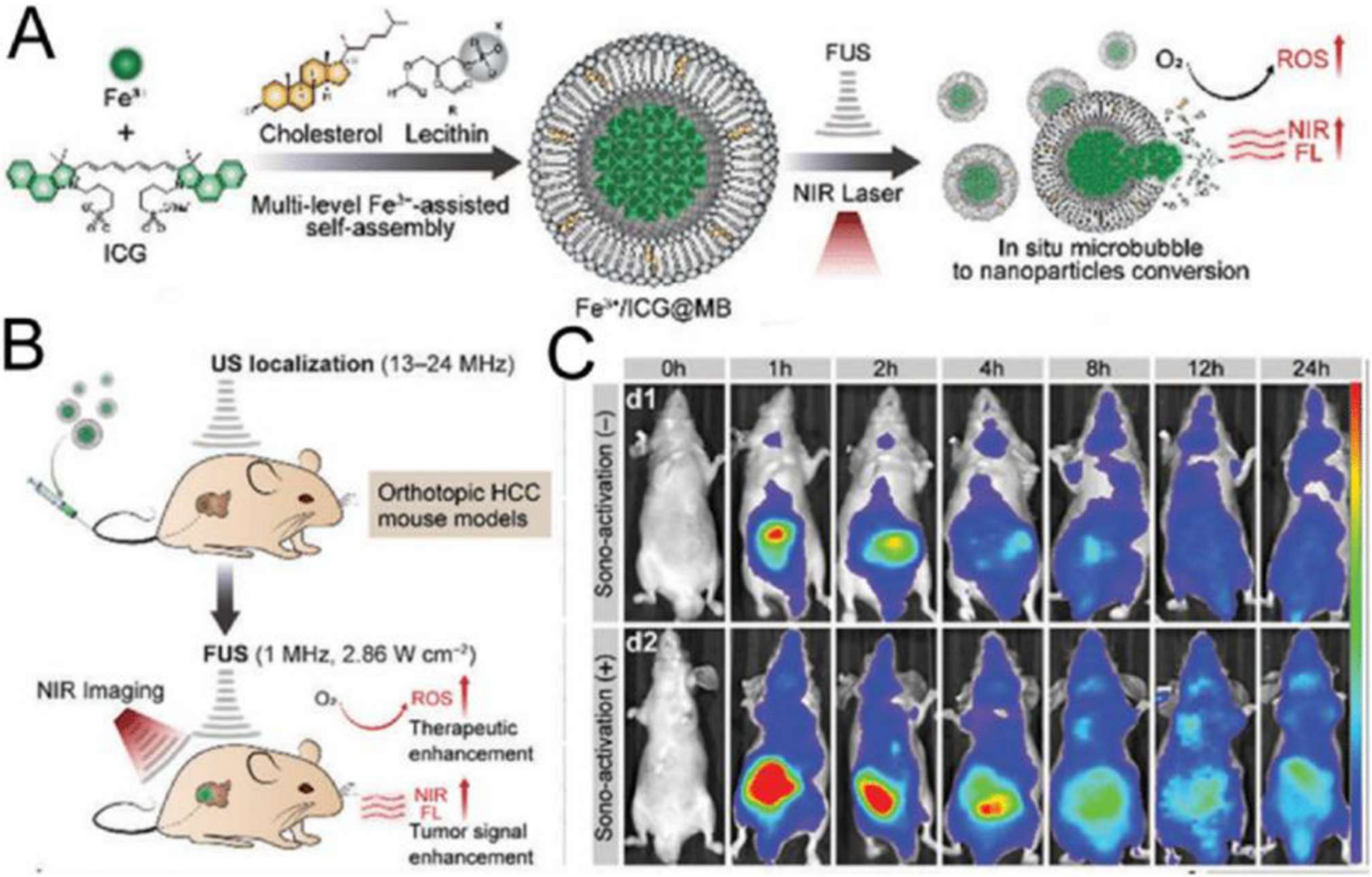






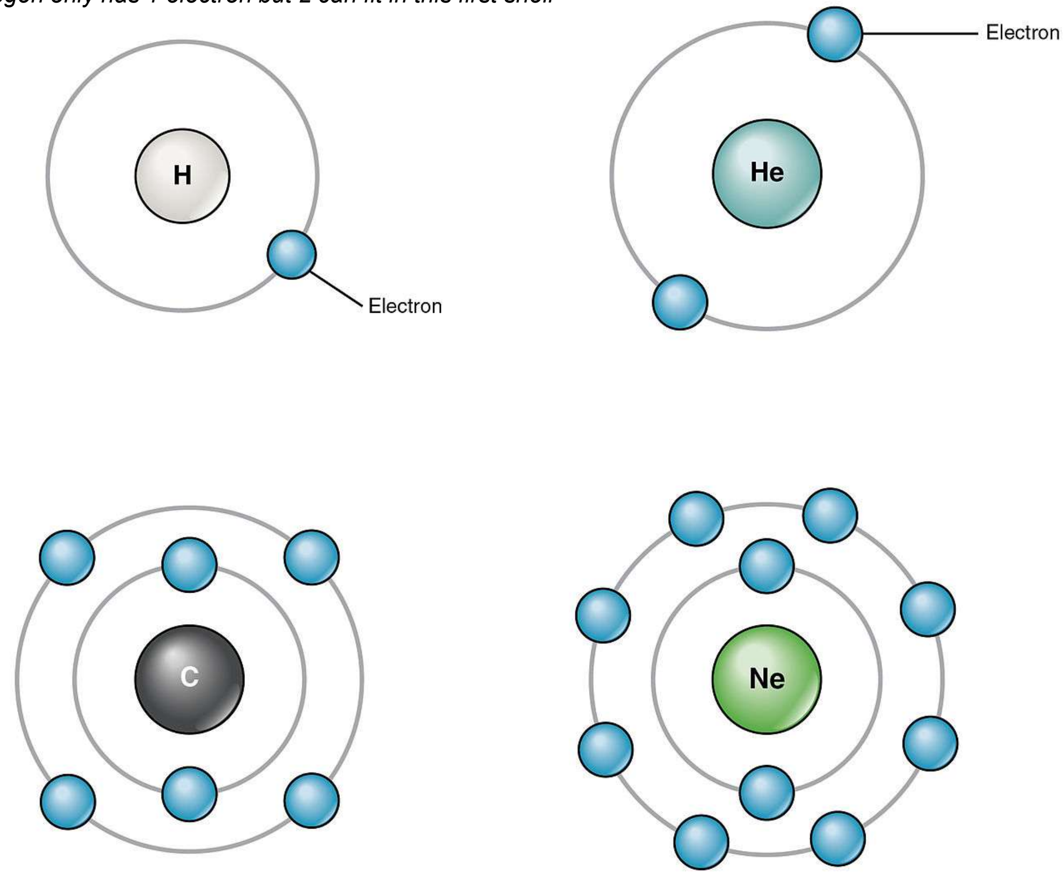
Imaging biomarkers of neurodegeneration. Coronal structural MRI section (panel **A**) and 18F-fluorodeoxyglucose (FDG) PET (panel **B**) from a patient with Alzheimer's disease (AD). *Int. J. Mol. Sci.* **2018**, *19*(12), 3702

Figure: https://commons.wikimedia.org/wiki/File:Radiation_Sickness.png


A
Fe³⁺
ICG
Cholesterol
Lecithin
Multi-level Fe³⁺-assisted self-assembly
Fe³⁺/ICG@MB
FUS
NIR Laser
O₂
ROS
NIR FL
In situ microbubble to nanoparticles conversion
B
US localization (13–24 MHz)
Orthotopic HCC mouse models
FUS (1 MHz, 2.86 W cm⁻²)
NIR imaging
Therapeutic enhancement
Tumor signal enhancement
C
0h
1h
2h
4h
8h
12h
24h
d1
d2
Sono-activation (–)
Sono-activation (+)

- **Electrons** (-) are distributed around the atomic nucleus in electron shells, where they are in constant motion
  - The first electron shell holds up to 2 electrons
  - The second electron shell holds up to 8 electrons
  - The third electron shell also holds up to 8 electrons

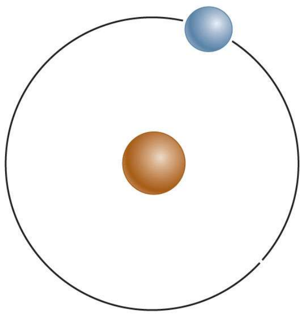
Hydrogen only has 1 electron but 2 can fit in this first shell



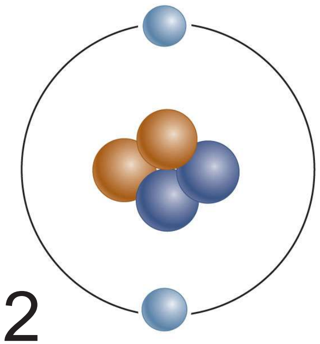
Carbon has 6 electrons, with 2 in the first shell and the remaining 4 in the second shell (which could hold up to 8)

Figure: https://commons.wikimedia.org/wiki/File:206_Electron_Shells-01.jpg

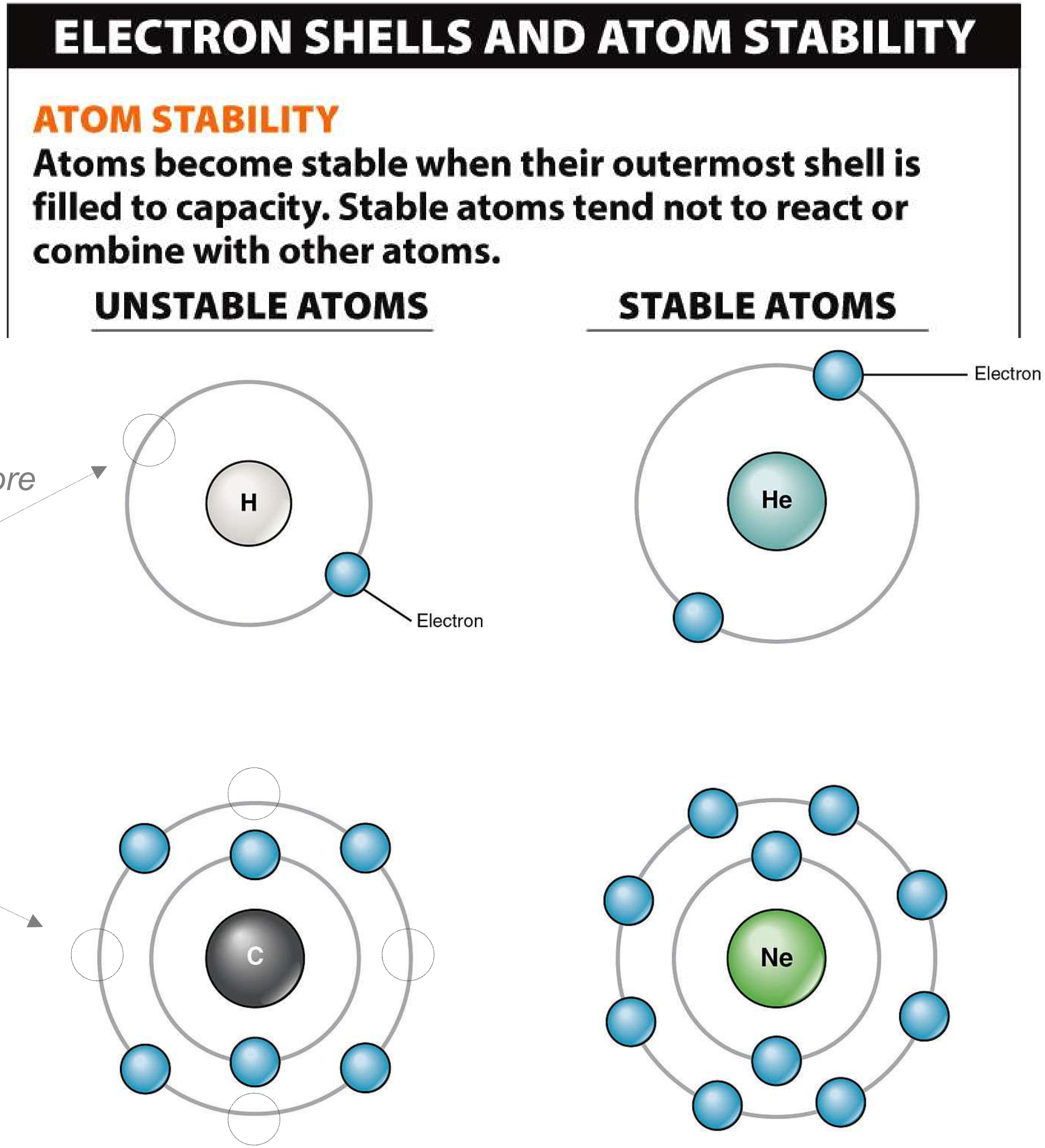
- Electrons are most stable when the outer shell of the atom is full
  - To fill their shells, atoms interact with other atoms, **losing**, **gaining**, or **sharing** electrons



Hydrogen (H)
- Atomic # = 1
- Outer shell not filled
- **Unstable**
- Will interact with other atoms



Helium (He)
- Atomic # = 2
- Outer shell filled
- **Stable**
- Will NOT interact with other atoms



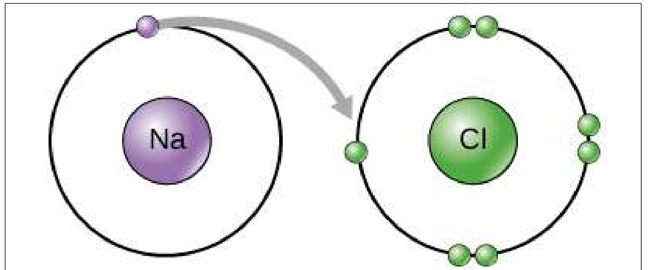
can hold 1 more

Only when atoms have electron vacancies in their outermost shell are they likely to interact with other atoms.

can hold 4 more

Figure: https://commons.wikimedia.org/wiki/File:206_Electron_Shells-01.jpg

- Atoms that lose or gain electrons to become stable are called **ions**



Sodium loses an electron: cation *Chloride gains an electron: anion*

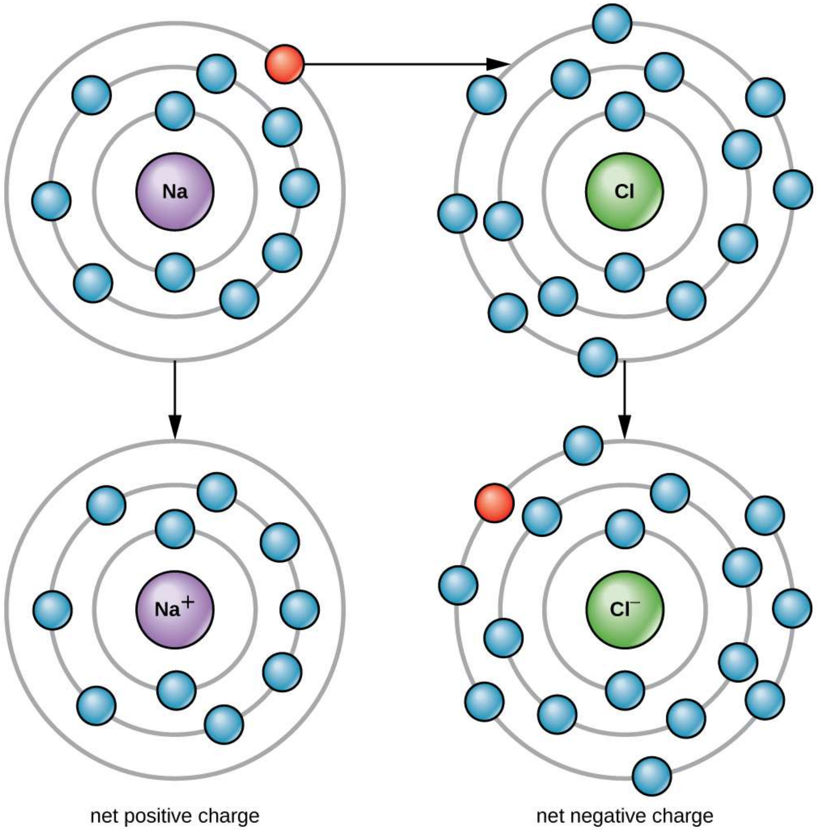
  - An atom that loses an electron (-) will have a positive charge = **cation**
    - e.g. carbon with 6 P+, 6 N, & 5 E-

  - An atom that gains an electron (-) will have a negative charge = **anion**
    - e.g. carbon with 6 P+, 6 N, & 7 E-

Figure: https://commons.wikimedia.org/wiki/File:Figure_02_01_10.jpg
Figure: https://commons.wikimedia.org/wiki/File:Lost_an_electron_joke.png

IONS ARE CHARGED ATOMS

An atom that loses one or more electrons becomes positively charged, while an atom that acquires electrons becomes negatively charged. This transfer of electrons is driven by the fact that atoms with full outer electron shells are more stable.



cation

SODIUM ION

- 11 Protons
- 11 Neutrons
- 10 Electrons

CHLORIDE ION

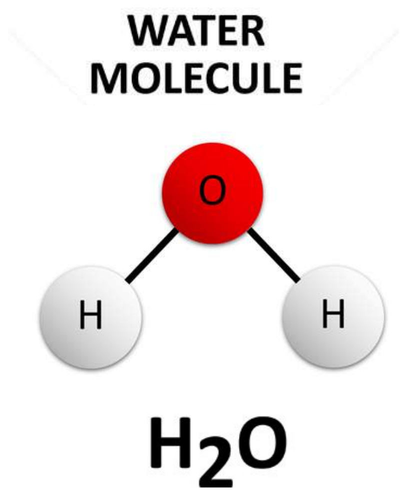
- 17 Protons
- 17 Neutrons
- 18 Electrons

anion

Figure: https://commons.wikimedia.org/wiki/File:OSC_Microbio_00_AA_bonds.jpg

Atoms link together using chemical bonds to form molecules

- The interaction of atoms can cause them to attach together through **chemical bonds**
  - 2 or more atoms joined together by chemical bonds = **molecule**
    - e.g. oxygen joins to 2 hydrogen atoms through chemical bonds to make a water molecule



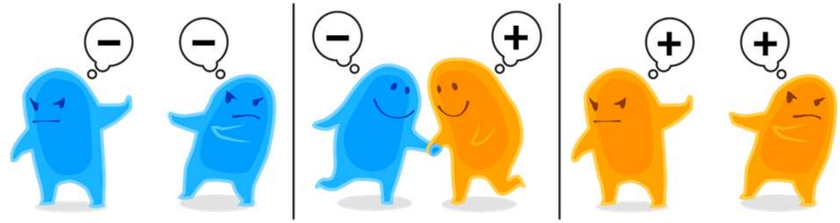
  - *Remember: life is organized – atoms make up **molecules**, which make up cells, which make up us*
  - *Remember: life requires energy & nutrients, which we get through eating **molecules** like fats & proteins*

Figure: https://commons.wikimedia.org/wiki/File:Water_Molecule_Ball_and_Stick.png

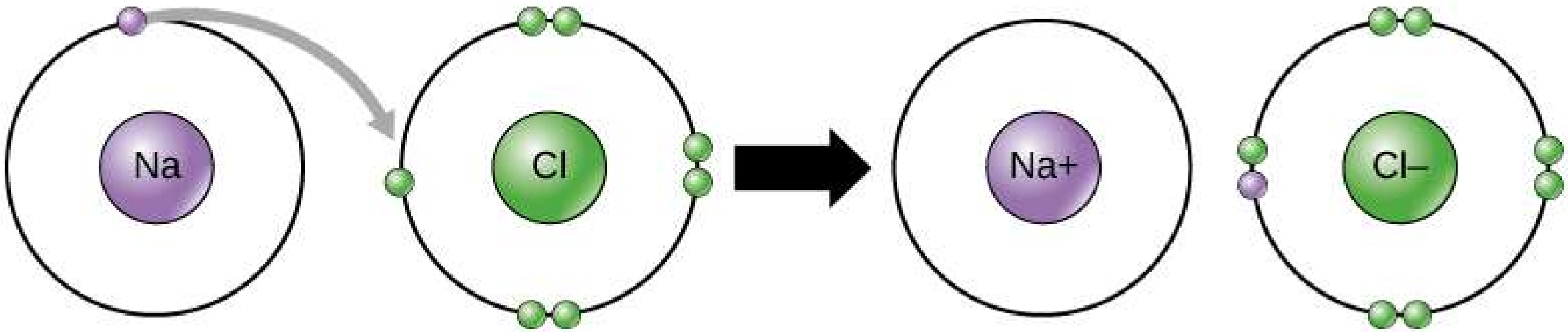
- There are many types of chemical bonds that hold molecules together
  - 3 types are important for biology: they hold together our bodies & the foods that we eat

Type of Bond	Chemical Interaction	Strength
Ionic	An electron is transferred, creating positive & negative ions that attract one another	Moderate – breaks easily in water
Covalent - Nonpolar - Polar	Electrons are shared between atoms - Equally - Unequally	Strong
Hydrogen	A positive atom within a molecule is attracted to a negative atom in a different molecule	Weak

- **Ionic Bonds**
  - The electrical attraction between cations (+) & anions (-) pulls atoms together
    - *similar to magnets*



cations (+) and anions (-) attract, forming ionic bonds

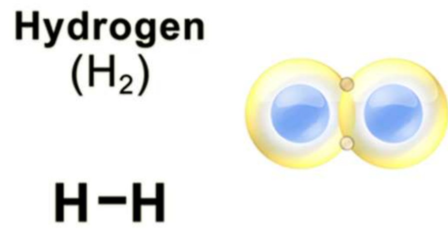


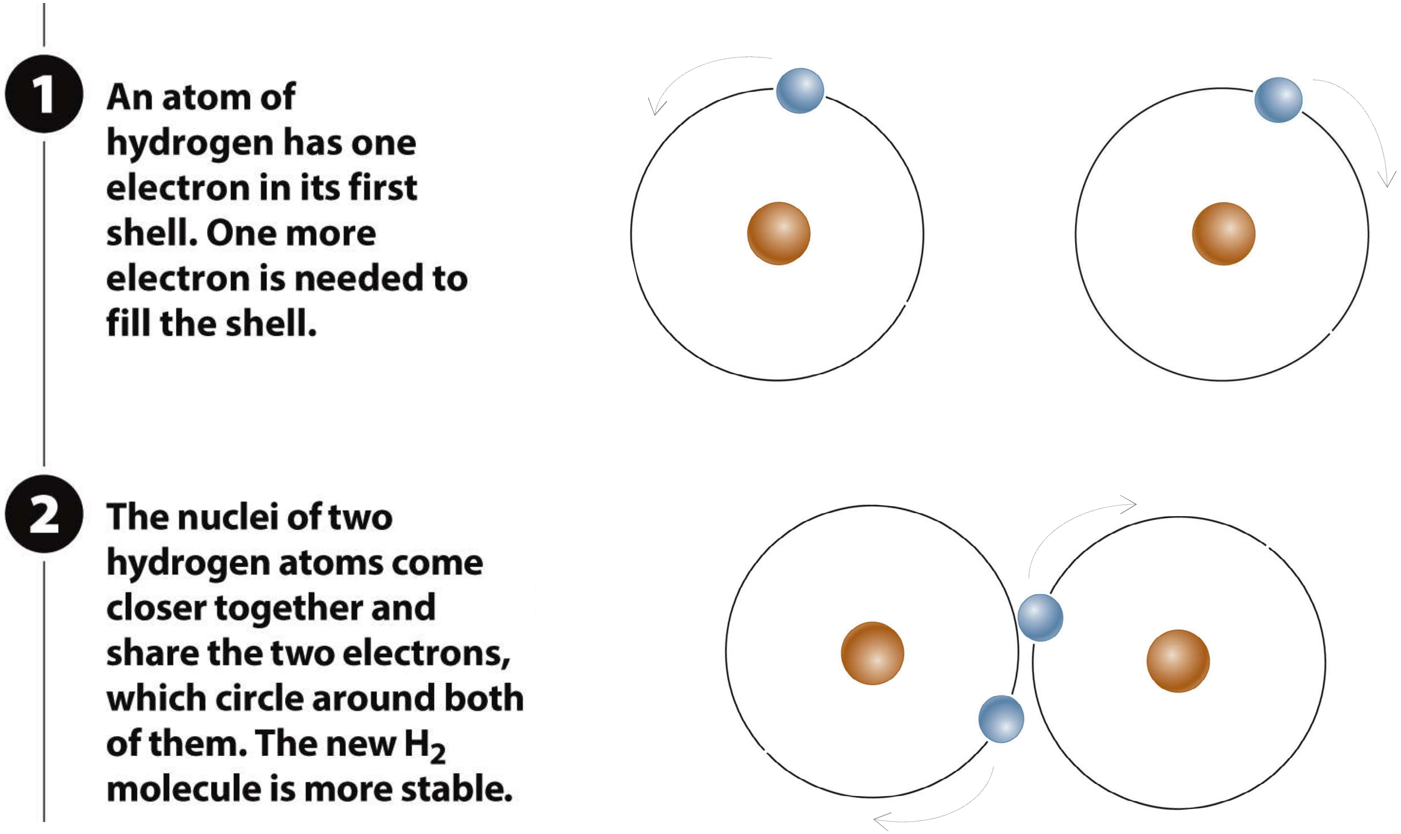
cation *anion* *ionic bond forms between them*

Figure: https://commons.wikimedia.org/wiki/File:Ion-attraction-snd.svg

Figure: https://commons.wikimedia.org/wiki/File:Figure_02_01_10.jpg

- **Covalent Bonds**
  - Two atoms <u>share</u> electrons
    - this means each shared electron will orbit its own atom, but also orbit the other atom



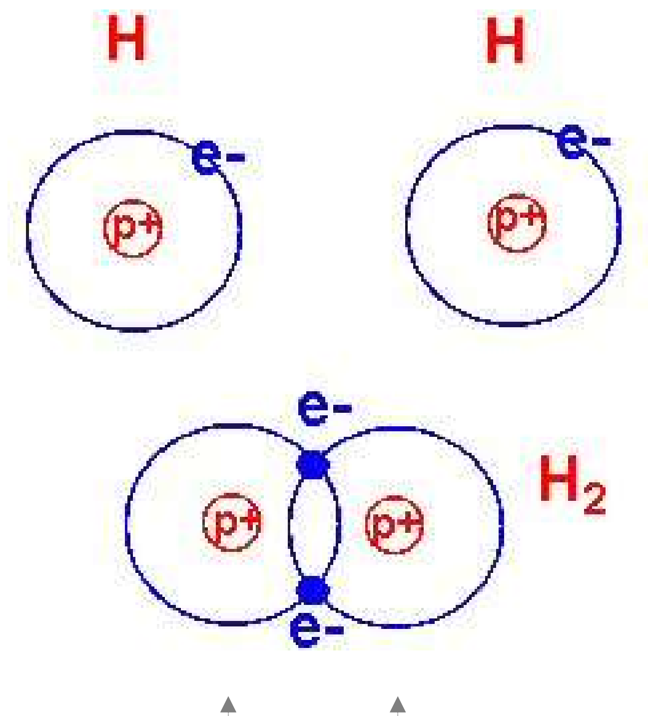


Remember, electrons are constantly orbiting in the shell.
These are 2 unstable atoms.

Now the shells overlap and each electron is orbiting both atoms.
This is now a stable molecule.

Figure: https://commons.wikimedia.org/wiki/File:Covalent_Bonds.png

- 2 types of covalent bonds:
  - **Nonpolar** = electrons shared evenly = uncharged
    - No charge builds up on either side

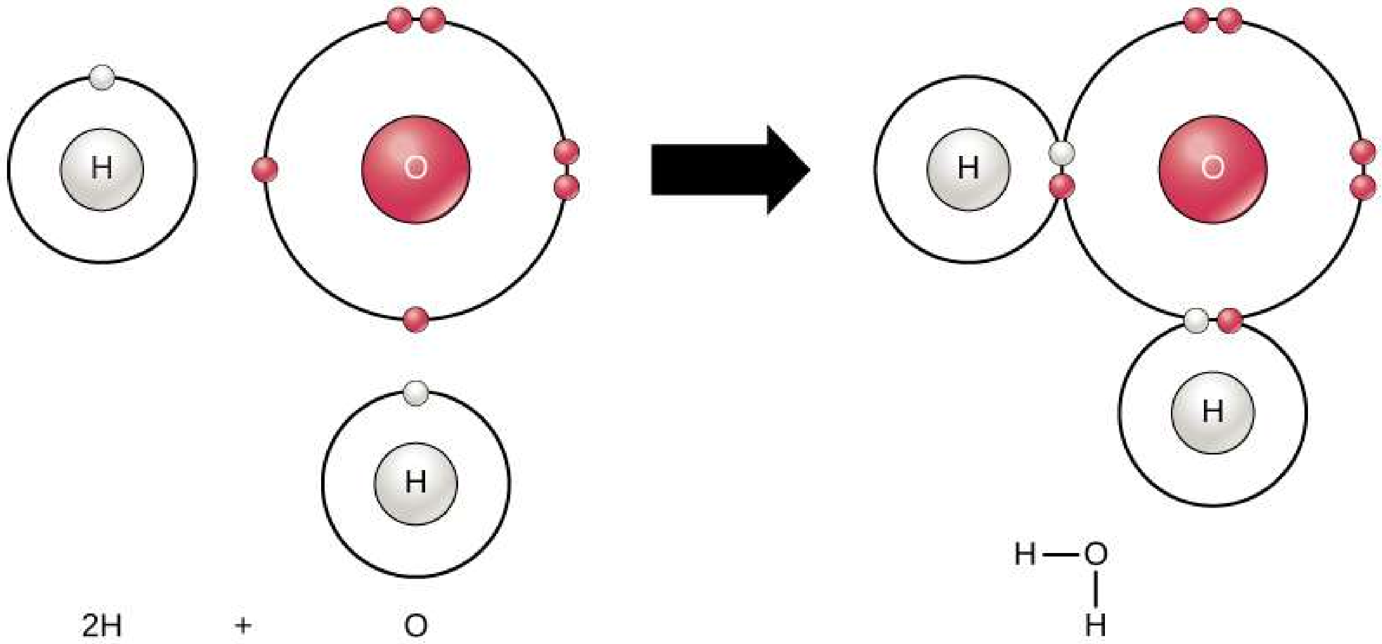


Same charge on either side (1 positive proton each), & electrons spend equal time near each nucleus

So no charge builds up on either side of this H_2 molecule: **nonpolar**

Figure: https://www.flickr.com/photos/loufcd/2811714713/

- 2 types of covalent bonds:
  - **Polar** = electrons NOT shared evenly = charged
    - A charge builds up on either side of the bond

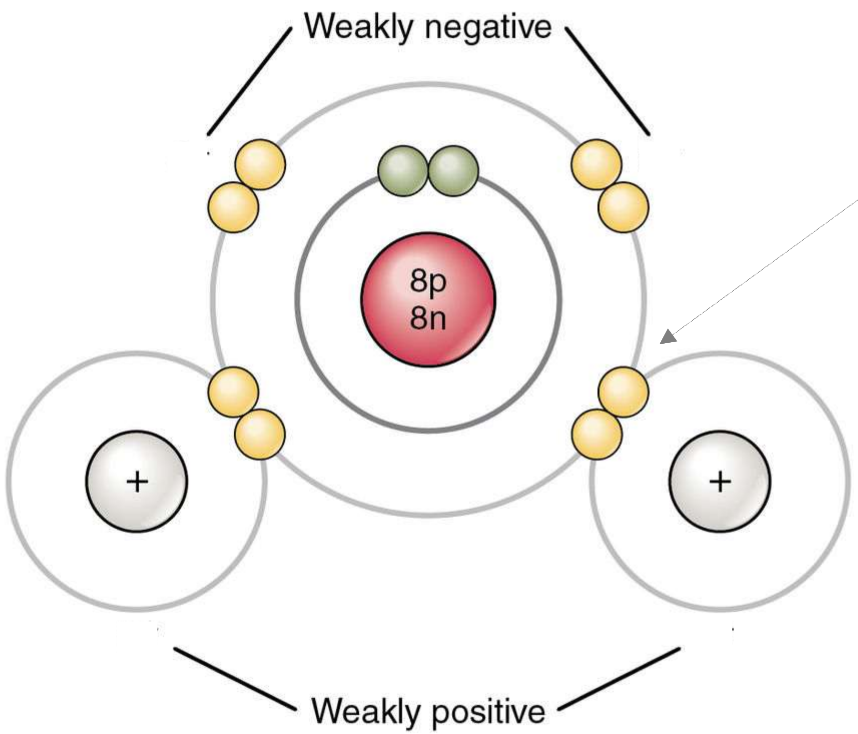


three unstable atoms

stable H_2O molecule

Figure: https://commons.wikimedia.org/wiki/File:Figure_02_01_08.jpg

- Polar = electrons NOT shared evenly = charged
  - Some atoms pull more on electrons than others *(based on "electronegativity," which is more detail than we will go into)*
    - Shared electrons now spend more time around one atom & less time around the other atom



Remember: electrons are in constant orbit but they're more often orbiting Oxygen here. That means the Hydrogen atoms have a positive proton but no negatives around them most of the time, so they're now kind of positive.

In a water molecule (H_2O) the oxygen pulls more on the electrons

Figure: https://commons.wikimedia.org/wiki/File:209_Polar_Covalent_Bonds_in_a_Water_Molecule.jpg

- Because oxygen pulls more on the electrons (-), which spend more time around it, it now has a slight negative charge
- The hydrogen atoms don't have electrons (-) around them very often, so now they are slightly positive

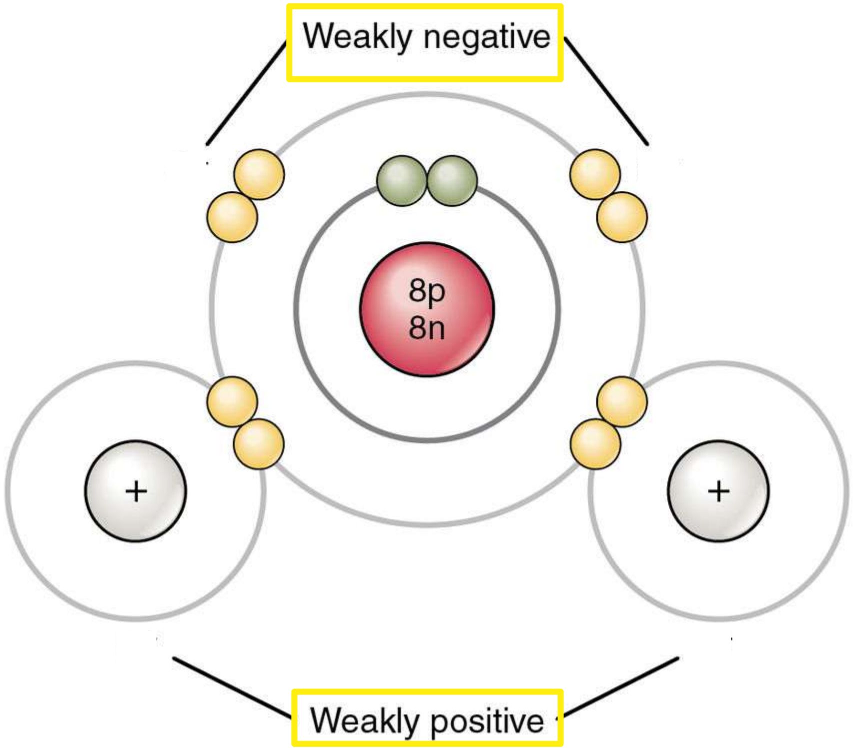
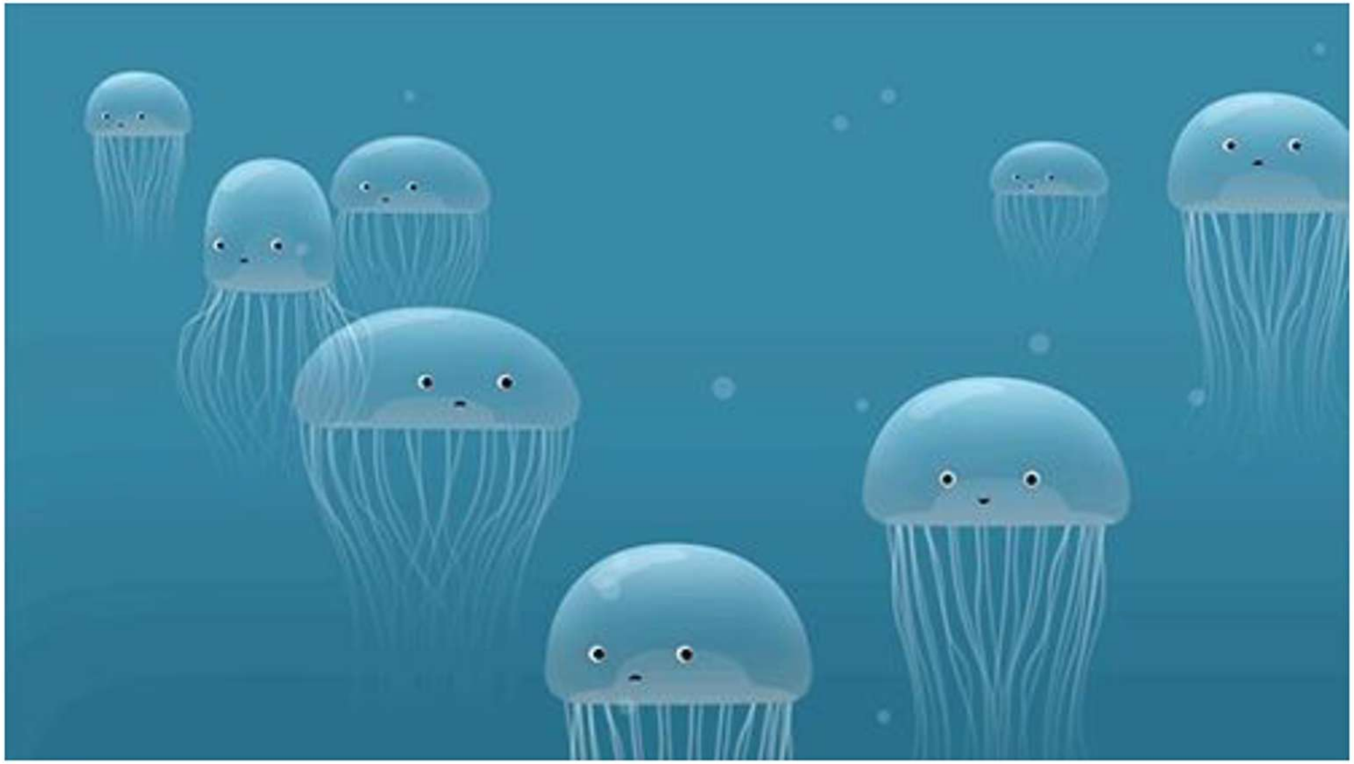


Figure: https://commons.wikimedia.org/wiki/File:209_Polar_Covalent_Bonds_in_a_Water_Molecule.jpg

Hydrogen bonds



The existence of living organisms relies on water.

Life is fundamentally dependent on hydrogen bonds.

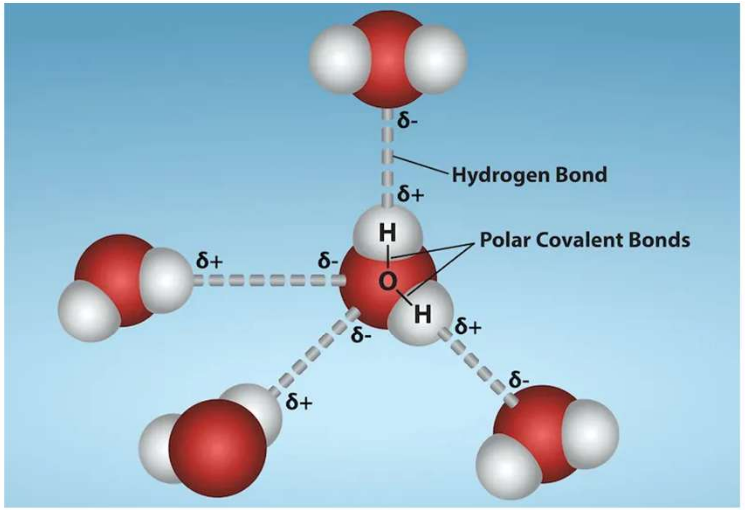
Figure: https://en.wikiversity.org/wiki/File:Water_molecules,_hydrogen_bonds.gif

Intermolecular forces
Hydrogen bond

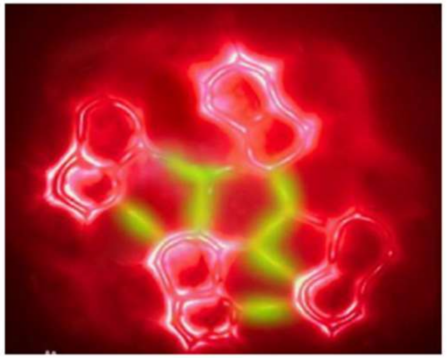
The hydrogen atom is covalently bonded to the electronegative atom X.

When it is in proximity to a small-radius electronegative atom Y (such as O, F, or N), a unique interaction occurs between X and Y, with H serving as a medium to facilitate the formation of an X-H...Y interaction known as a hydrogen bond.

X and Y may consist of identical atoms, exemplified by the hydrogen bonds present between water molecules; alternatively, they can be different types of atoms, such as ammonia hydrate ($NH_3 \cdot H_2O$).



The pair of electrons shared between the O and H atoms is predominantly localized in the region between the two atoms. Given that the H atom possesses only this single pair of electrons, it can be likened to a half-peeled orange, with a significant portion of its nucleus exposed, resulting in a certain degree of positive charge. In contrast, oxygen, with its eight electrons, exhibits electronegativity to some extent.
When an O atom from one water molecule approaches a H atom from another water molecule, an electrostatic attraction occurs; this interaction is referred to as a hydrogen bond.

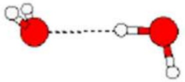


Professor Qiu Xiaohui:

A hydrogen bond is like two people holding hands; they can easily pull apart or let go.

On the other hand, a chemical bond is more like their own hands and feet being tied together, making it impossible to separate.

Figure: https://en.wikiversity.org/wiki/File:Water_molecules,_hydrogen_bonds.gif



The 6 principles governing hydrogen bonding:

1. The formation of hydrogen bonds primarily arises from electrostatic forces. A partial covalent bond is established between the hydrogen atom (H) and the electronegative atom (Y) due to electrostatic interactions resulting from charge transfer between donor and acceptor species. The creation of this covalent bond is a consequence of discrete actions.

2. The typical covalent bond formed between atom X and hydrogen (X–H) is characterized as a polar bond. The strength of the interaction H...Y increases with the increasing electronegativity of atom X.

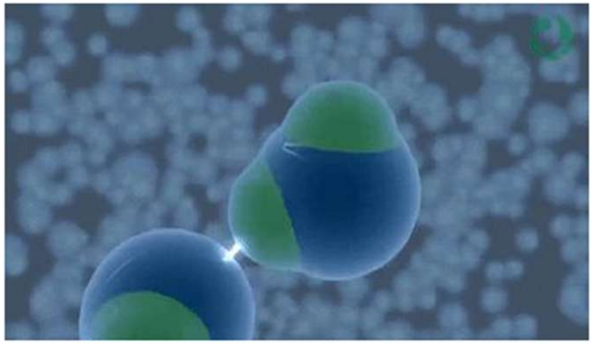
3. The dihedral angle formed by X—H...Y approaches linearity, ideally close to 180 degrees; stronger hydrogen bonds correspond to shorter distances between H and Y.

4. Hydrogen bonding causes an increase in the X-H distance, and the longer the X-H...Y bond length, the stronger the H...Y hydrogen bond will be, and some new vibrational modes will also be formed.

5. Hydrogen bonds also generate distinctive NMR signals, including the spin coupling interactions between atoms X and Y resulting from hydrogen bonding.

6. The Gibbs free energy associated with these hydrogen bonds exceeds that of thermal energy within the system.

Figure: https://en.wikiversity.org/wiki/File:Water_molecules,_hydrogen_bonds.gif



The average bond energy data for common hydrogen bonds is:

F—H … F (155 kJ/mol or 40 kcal/mol)

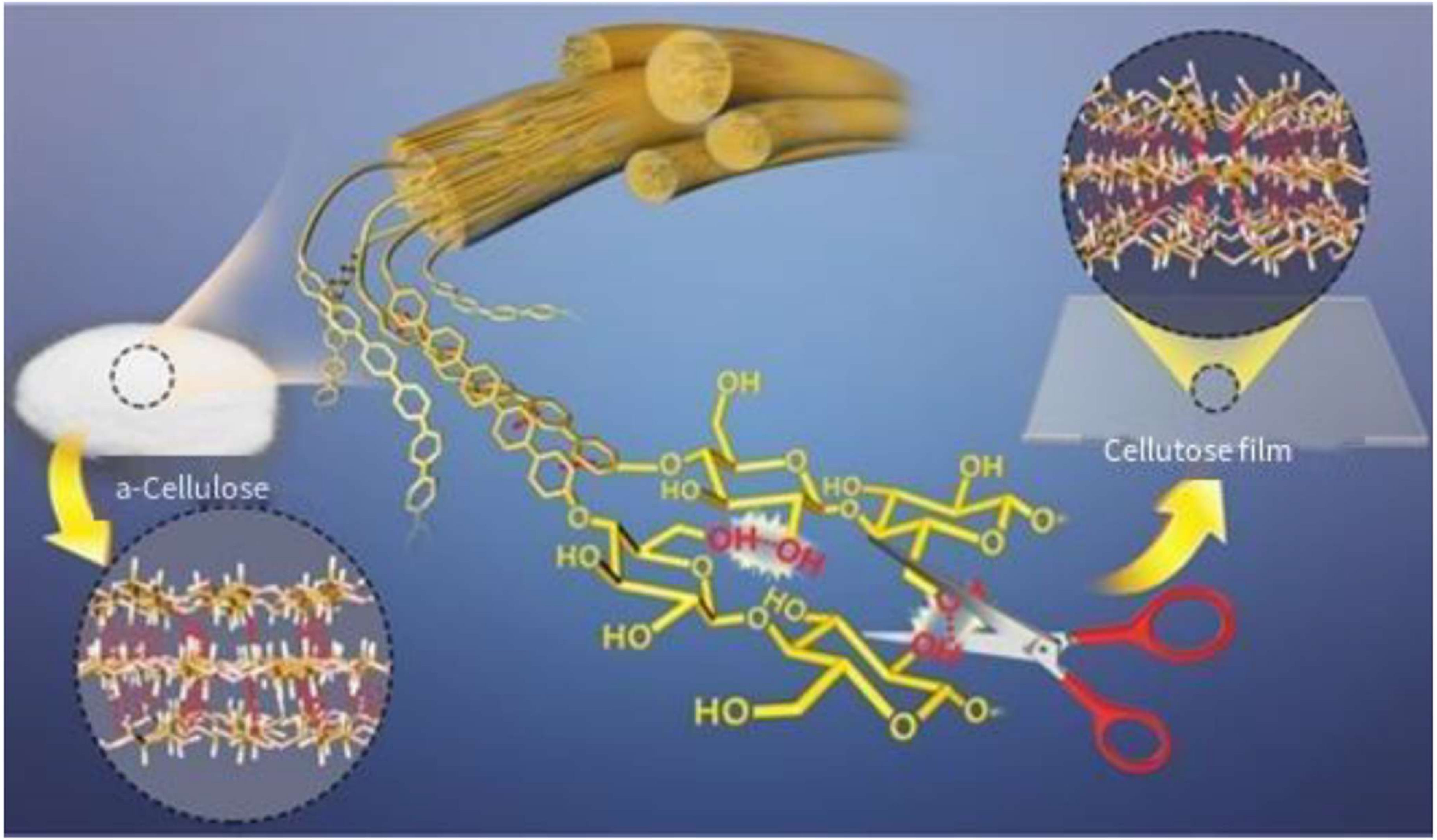
O—H … N (29 kJ/mol or 6.9 kcal/mol)

O—H … O (21 kJ/mol or 5.0 kcal/mol)

N—H … N (13 kJ/mol or 3.1 kcal/mol)

N—H … O (8 kJ/mol or 1.9 kcal/mol)

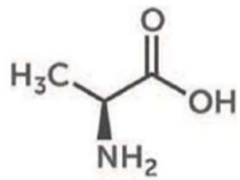
https://taylorsciencegeeks.weebly.com/blog/archives/02-2019/5


a-Cellulose
Cellutose film
OH
HO
O
OH
HO
OH OH
HO
HO
HO
HO
O

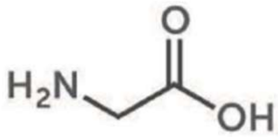
THE CHEMISTRY OF SPIDERWEBS

October is mating season for some spiders. Like them or loathe them, these arachnids use some fascinating biochemistry to spin webs with unique material properties that scientists want to emulate.

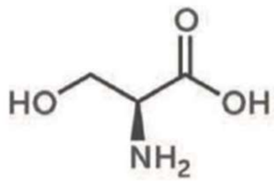
SPIDER SILK'S ELASTICITY AND STRENGTH



ALANINE

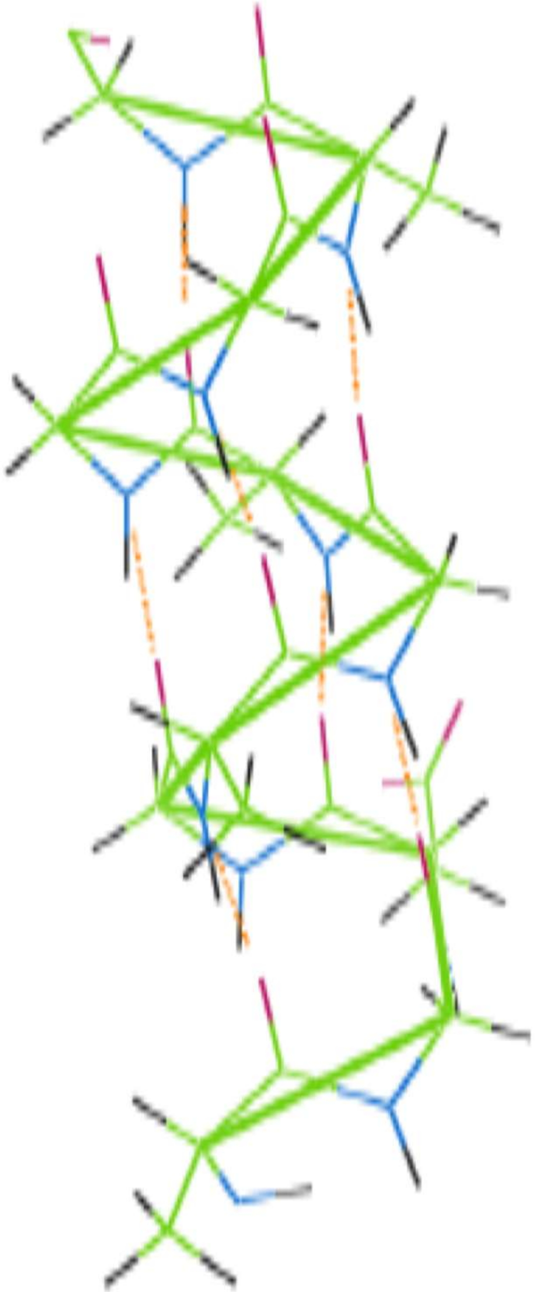


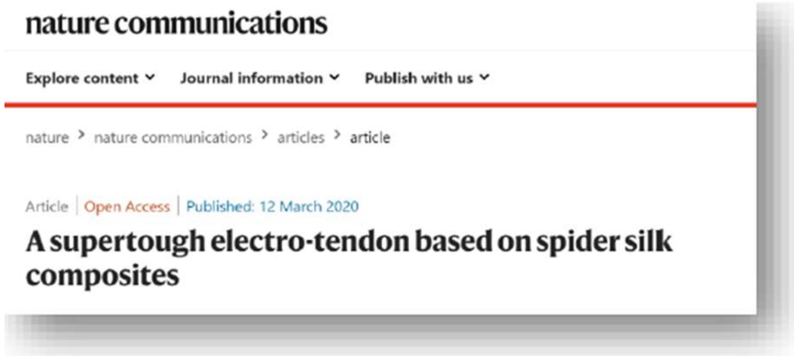
GLYCINE



SERINE

Spider silk is a protein fiber. Major amino acids in the silk proteins are alanine and glycine. Serine and proline are also present in significant quantities in some types of silk. Glycine-rich regions give spider silk its elasticity, forming amorphous areas in its structure. Alanine-rich regions link together through hydrogen bonds to form crystalline areas that give spider silk its strength.

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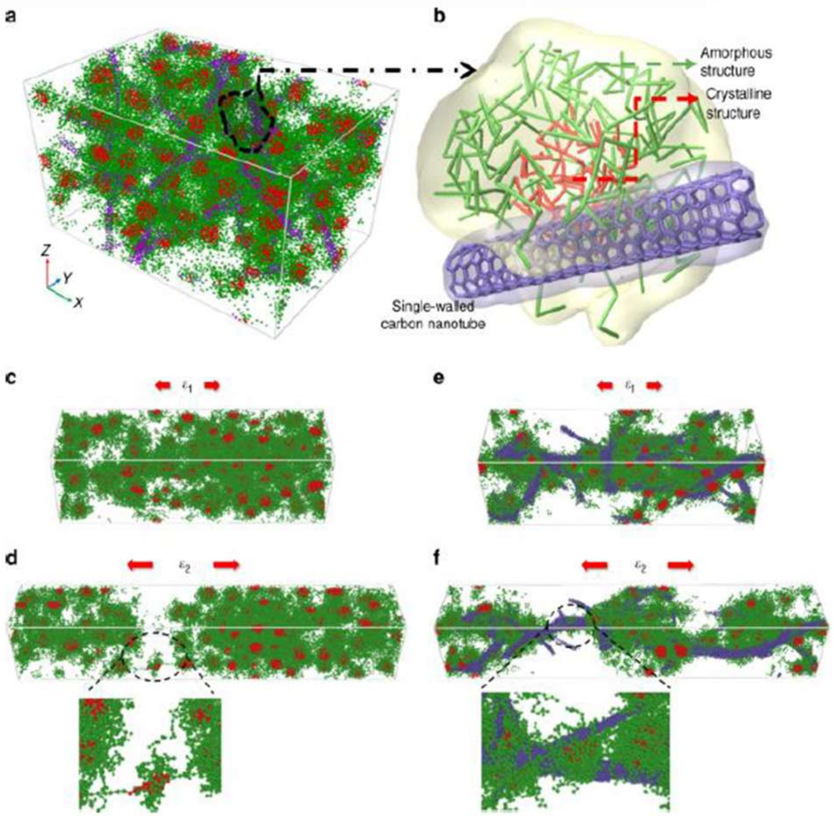
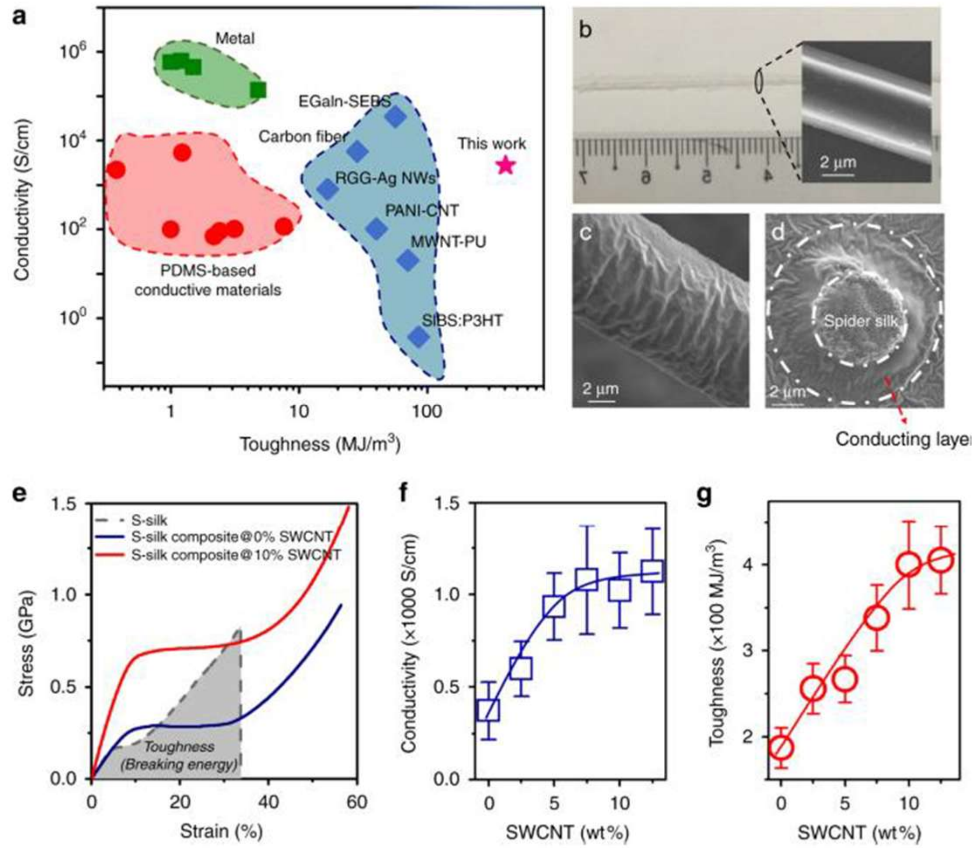
Article | Open Access | Published: 12 March 2020

A supertough electro-tendon based on spider silk composites

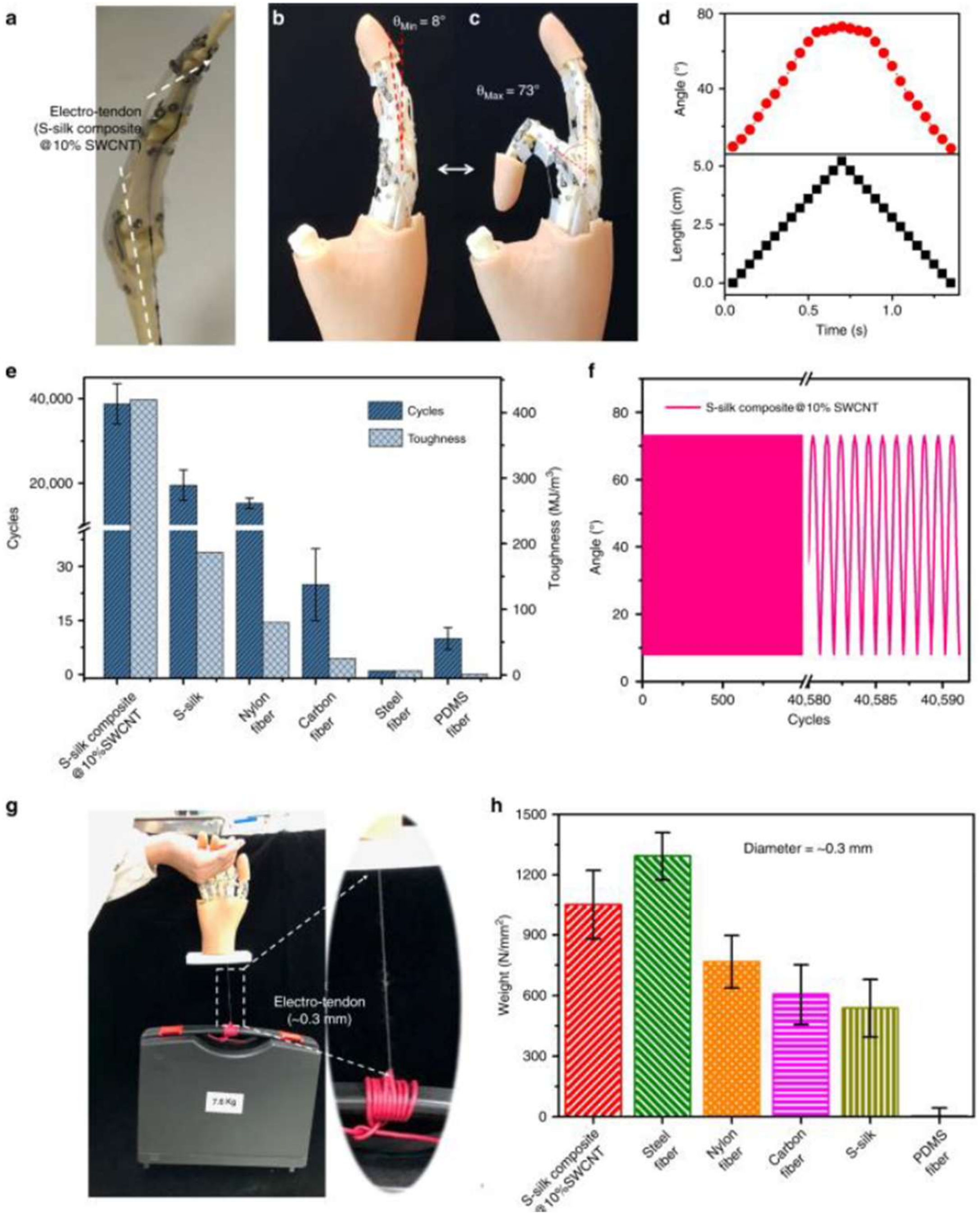


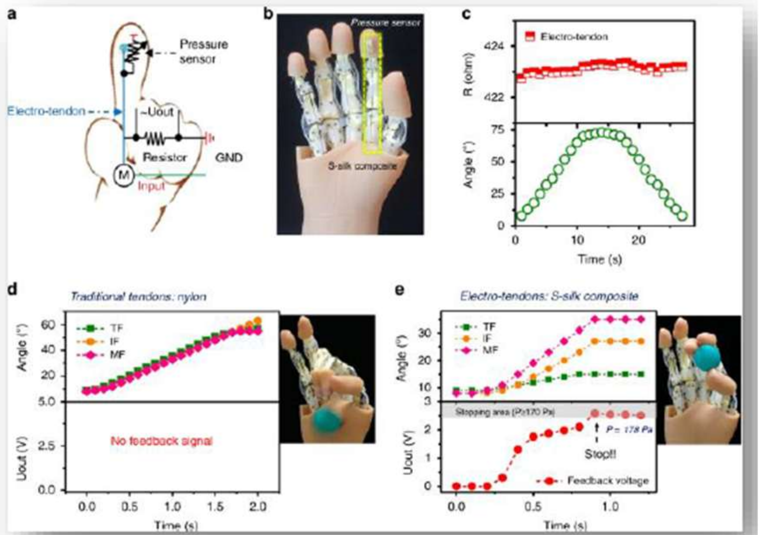


A typical mechanical test of the silk-SWCNT nanocomposite in DPD simulation.

Toughness and conductivity of spider silk composites.

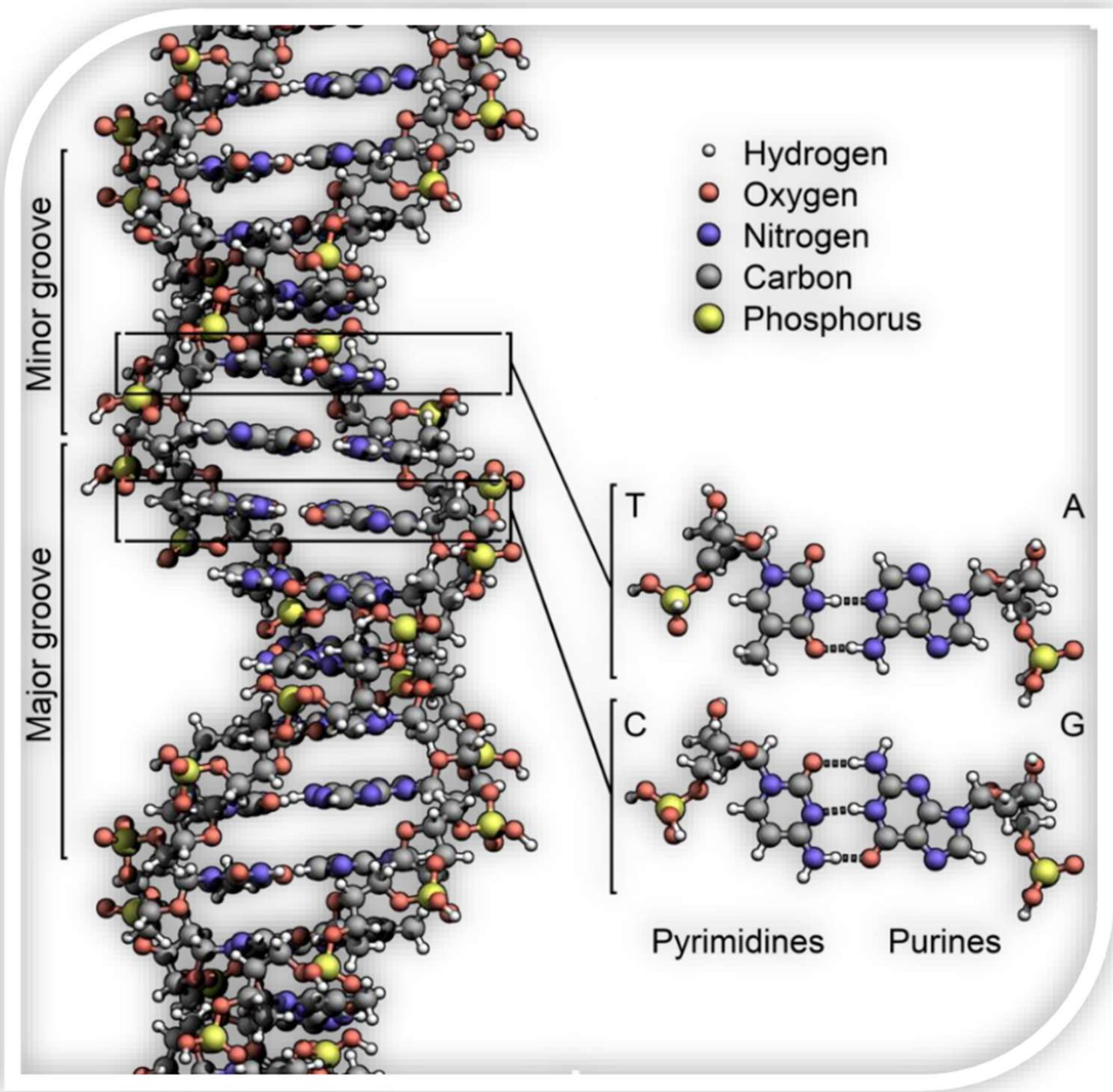
Applications of LHS



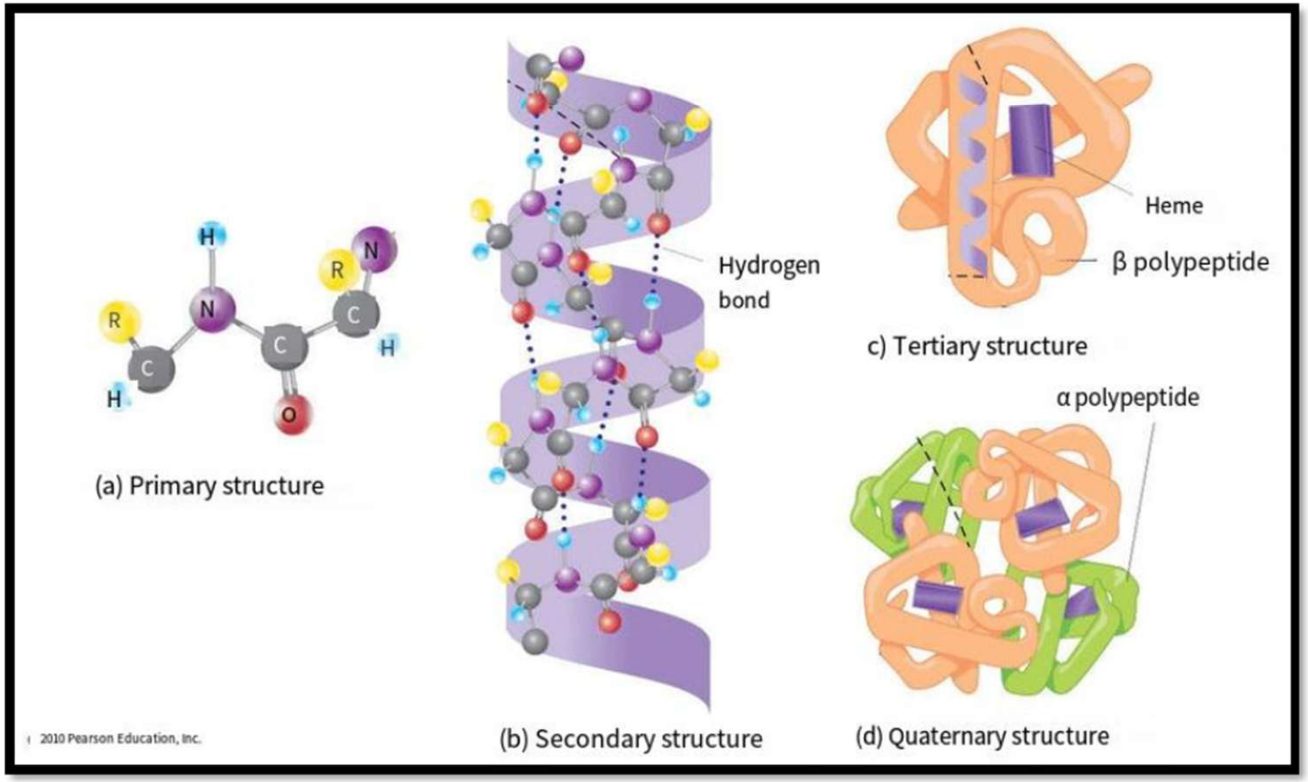


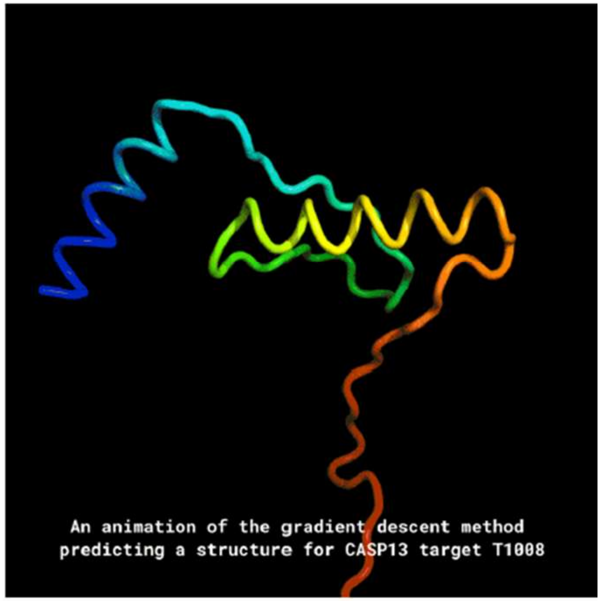
Performance of humanoid robotic hands assembled with S-silk composite as electro-tendon.

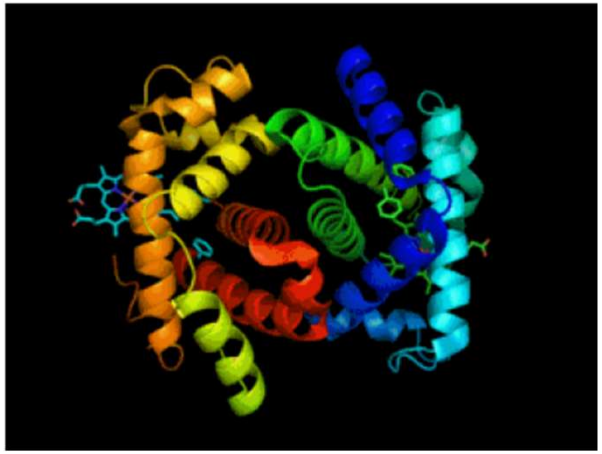
The significance of hydrogen bonds for life

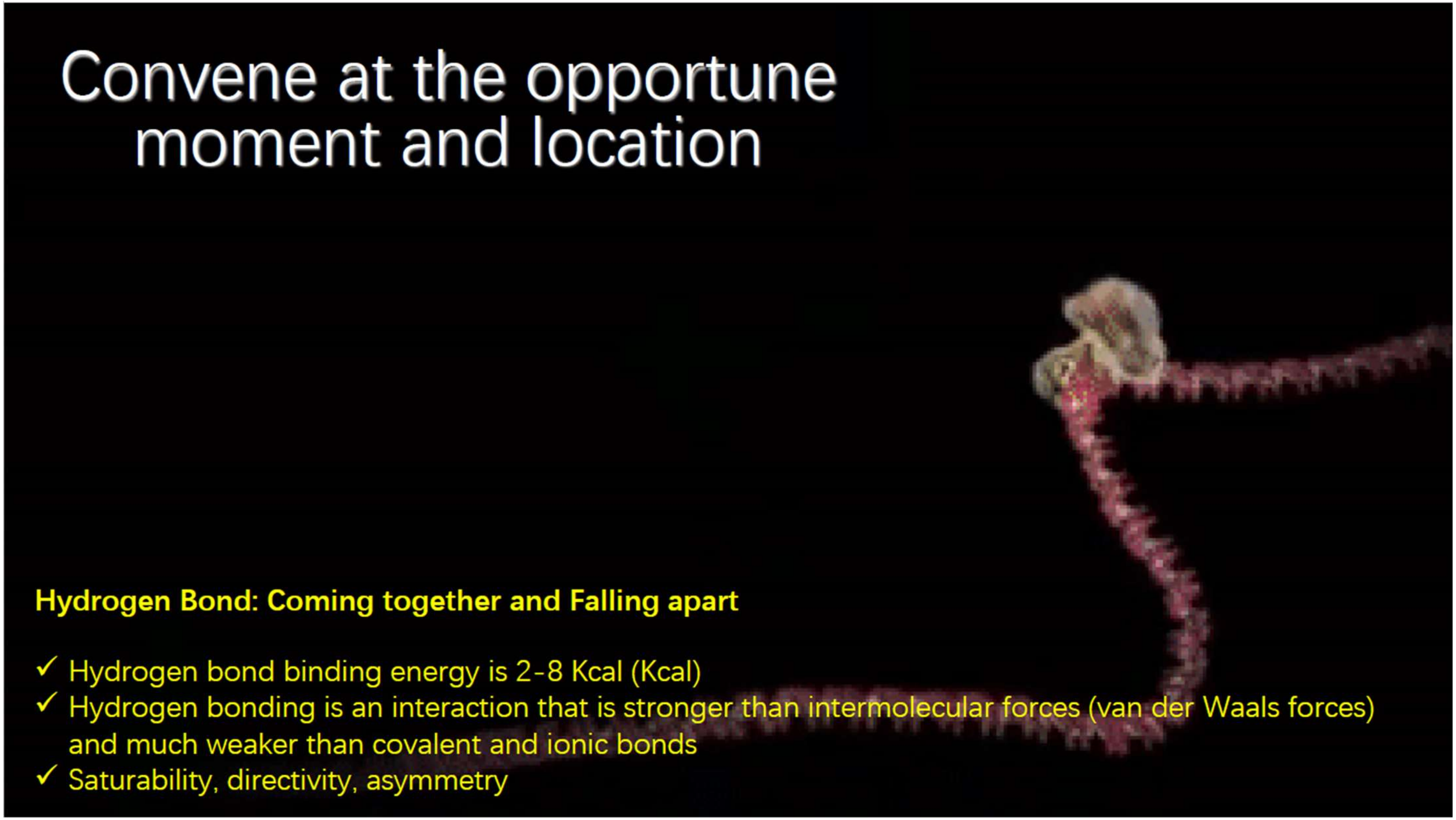


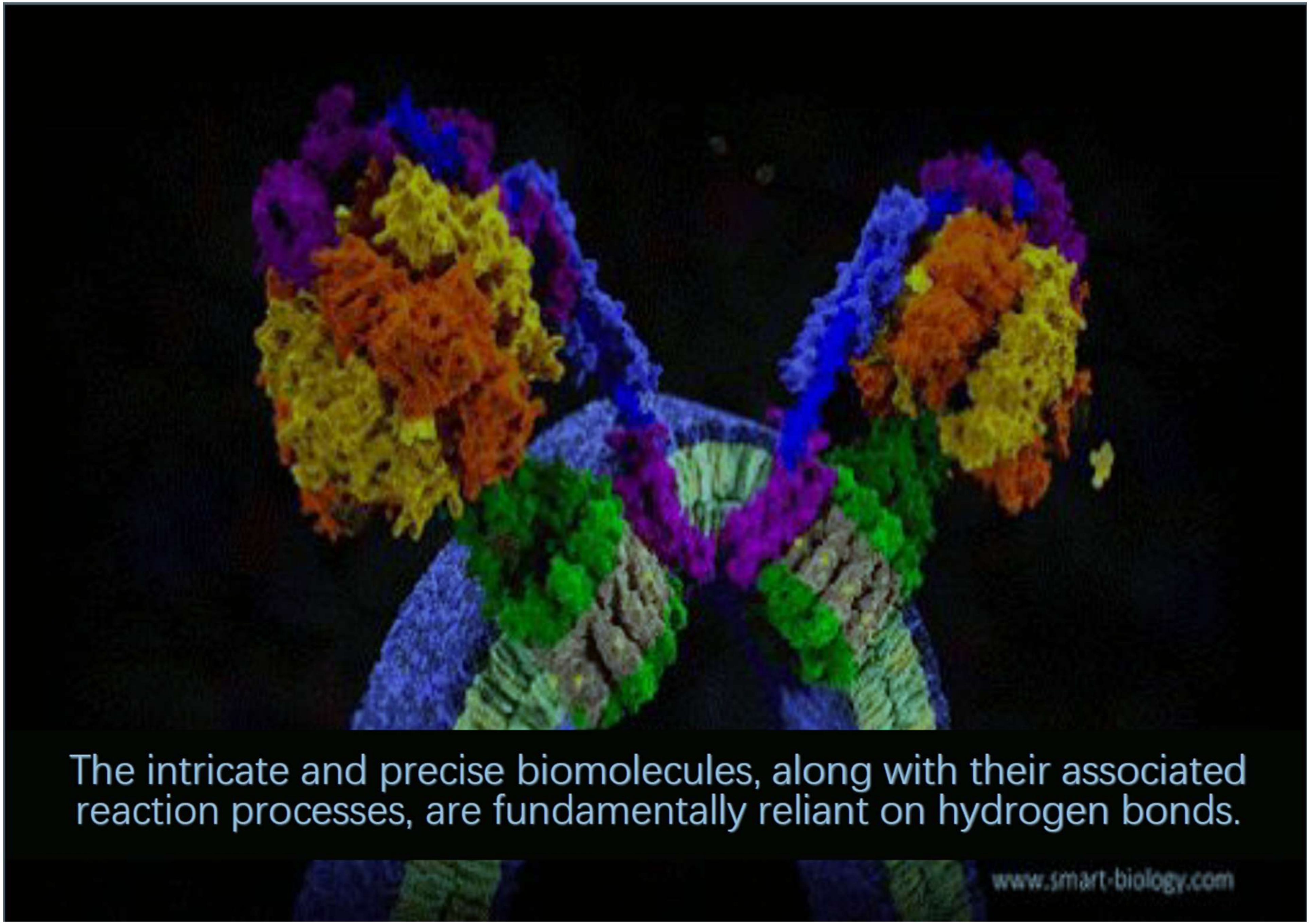
Hydrogen
Oxygen
Nitrogen
Carbon
Phosphorus
Minor groove
Major groove
T
A
C
G
Pyrimidines
Purines


Hydrogen bond
Heme
β polypeptide
c) Tertiary structure
α polypeptide
(a) Primary structure
(b) Secondary structure
(d) Quaternary structure
2010 Pearson Education, Inc.


An animation of the gradient descent method predicting a structure for CASP13 target T1008



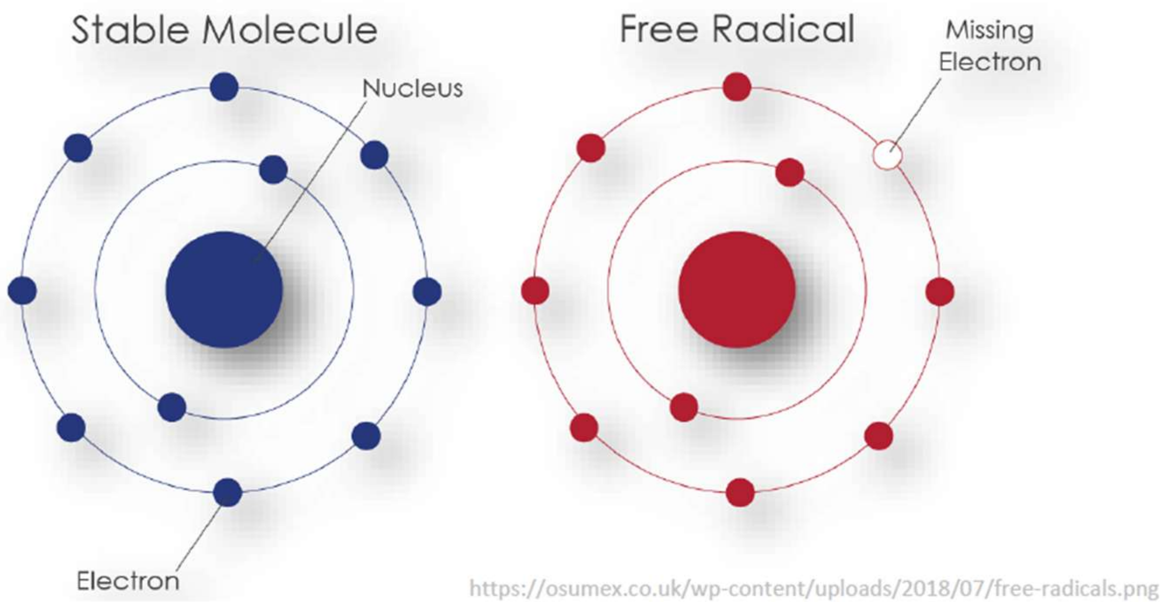

Convene at the opportune moment and location
Hydrogen Bond: Coming together and Falling apart
✓ Hydrogen bond binding energy is 2-8 Kcal (Kcal)
✓ Hydrogen bonding is an interaction that is stronger than intermolecular forces (van der Waals forces) and much weaker than covalent and ionic bonds
✓ Saturability, directivity, asymmetry


The intricate and precise biomolecules, along with their associated reaction processes, are fundamentally reliant on hydrogen bonds.
www.smart-biology.com

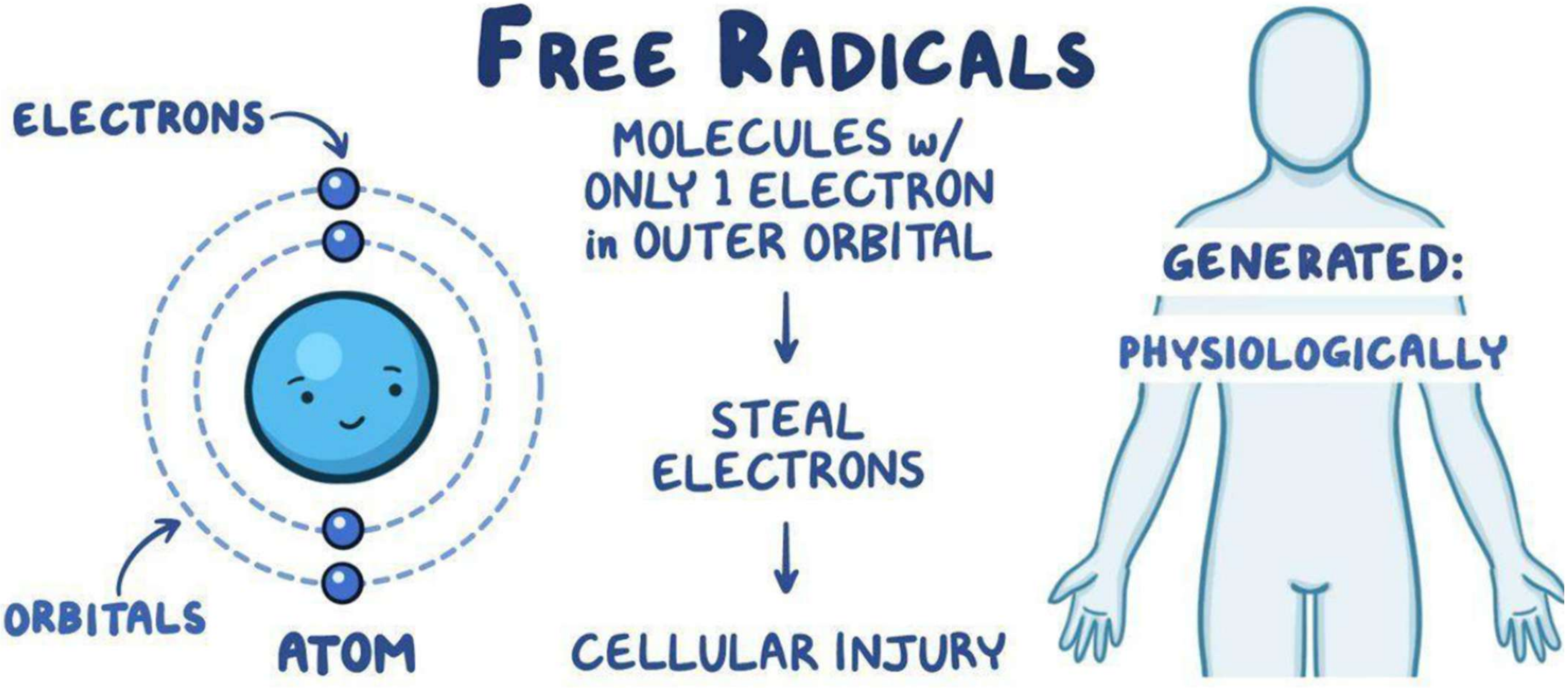
Free Radical

- Atoms interact with each other in order to stabilize
  - Some chemical reactions cause an atom or molecule to be stripped of its electrons
    - Can be caused by radiation, UV light, some chemicals & food additives, pesticides, pollution, cigarette smoke, etc.

An atom or molecule that has been stripped of its electrons & is highly unstable = free radical



- **Free radicals** are HIGHLY unstable: grab electrons from any nearby source to stabilize themselves
  - Stealing electrons = new free radicals, plus it damages the molecules so they can't do their jobs in the body anymore



https://d16qt3wv6xm098.cloudfront.net/NH5khCmqQiyorEyBcl0d-GLOSNy8eq-T/_.jpg

Figure: https://en.wikiversity.org/wiki/File:Water_molecules,_hydrogen_bonds.gif

- Free radicals are more & more common as pesticides, pollution, etc. increase in our environment
- <u>How can we fight free radicals?</u>
  - – There are special molecules that can stabilize free radicals without becoming free radicals themselves = **antioxidants**
    - break the chain reaction

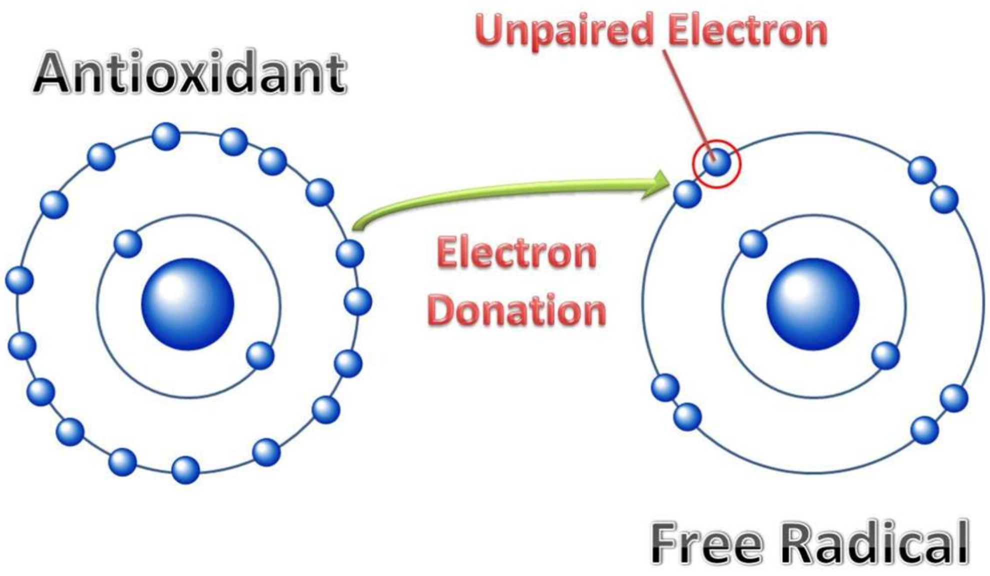


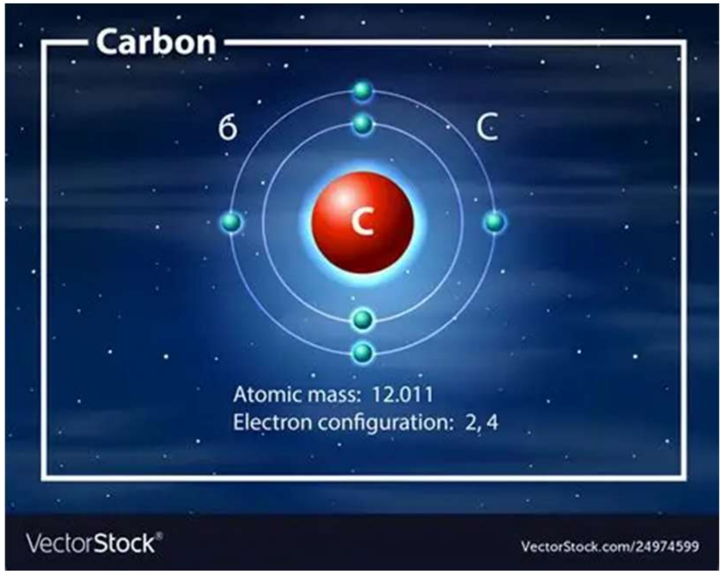
Figure: https://commons.wikimedia.org/wiki/File:Antioxidants_Free_radicals_Lobo_et_al.,.png

- Antioxidants include vitamins A, E, & C, flavenoids, carotenoids, etc.
  - Found in high amounts in fruits & vegetables (especially raw)
  - Moderate amounts in nuts, tea, & coffee
  - Low amounts in red wine & dark chocolate
    - *watch out for milk proteins in chocolate or coffee though – bind to & negate antioxidants*

https://pixabay.com/photos/antioxidant-apple-avocado-5954193/ https://commons.wikimedia.org/wiki/File:Dark_chocolate_Blanxart.jpg
https://commons.wikimedia.org/wiki/File:Mint_tea_-_Petr_Kratochvil.jpg

Carbon

- The carbon atom has unique properties that allow it to form covalent bonds to as many as four different atoms, making this versatile element ideal to serve as the basic structural component, or "backbone" of the macromolecules.



- Many complex molecules called macromolecules, such as proteins, nucleic acids (RNA and DNA), carbohydrates, and lipids. These macromolecules are a subset of organic molecules (any carbon-containing liquid, solid, or gas) that are especially important for life. The fundamental component for all of these macromolecules is carbon.

Figure: https://commons.wikimedia.org/wiki/File:Hot_Watering_Hole_Action.jpg

Hydrocarbons

- Isomers

Molecules can vary by the way atoms are arranged in three-dimensional space. The same combination of atoms occupies space in different ways. Isomers are structures with the same molecular formula and chemical composition but different arrangements of atoms in space.

- Enantiomers

Molecules that share the same chemical structure and chemical bonds but differ in the three-dimensional placement of atoms so that they are non-superimposable mirror images

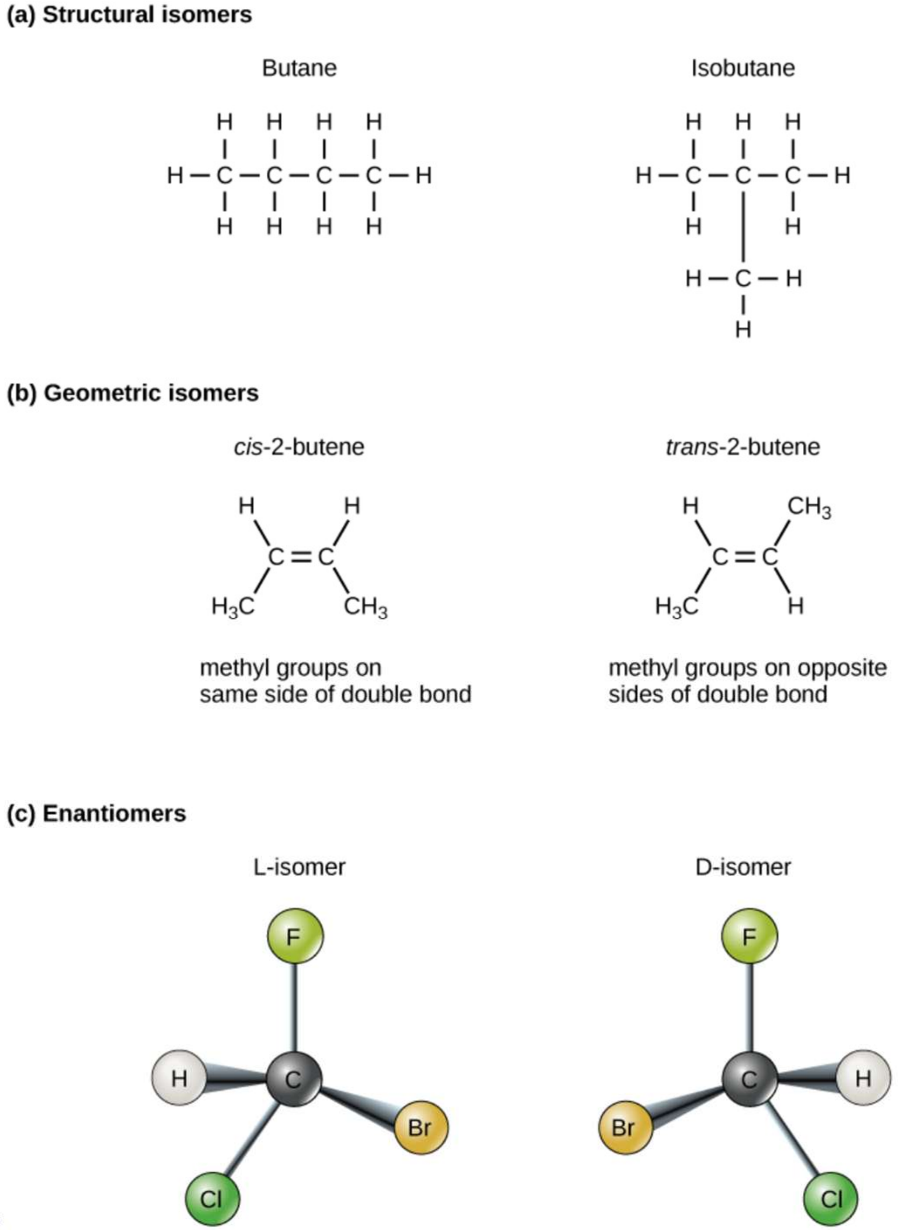


Figure: https://commons.wikimedia.org/wiki/File:Hot_Watering_Hole_Action.jpg

Hydrocarbons

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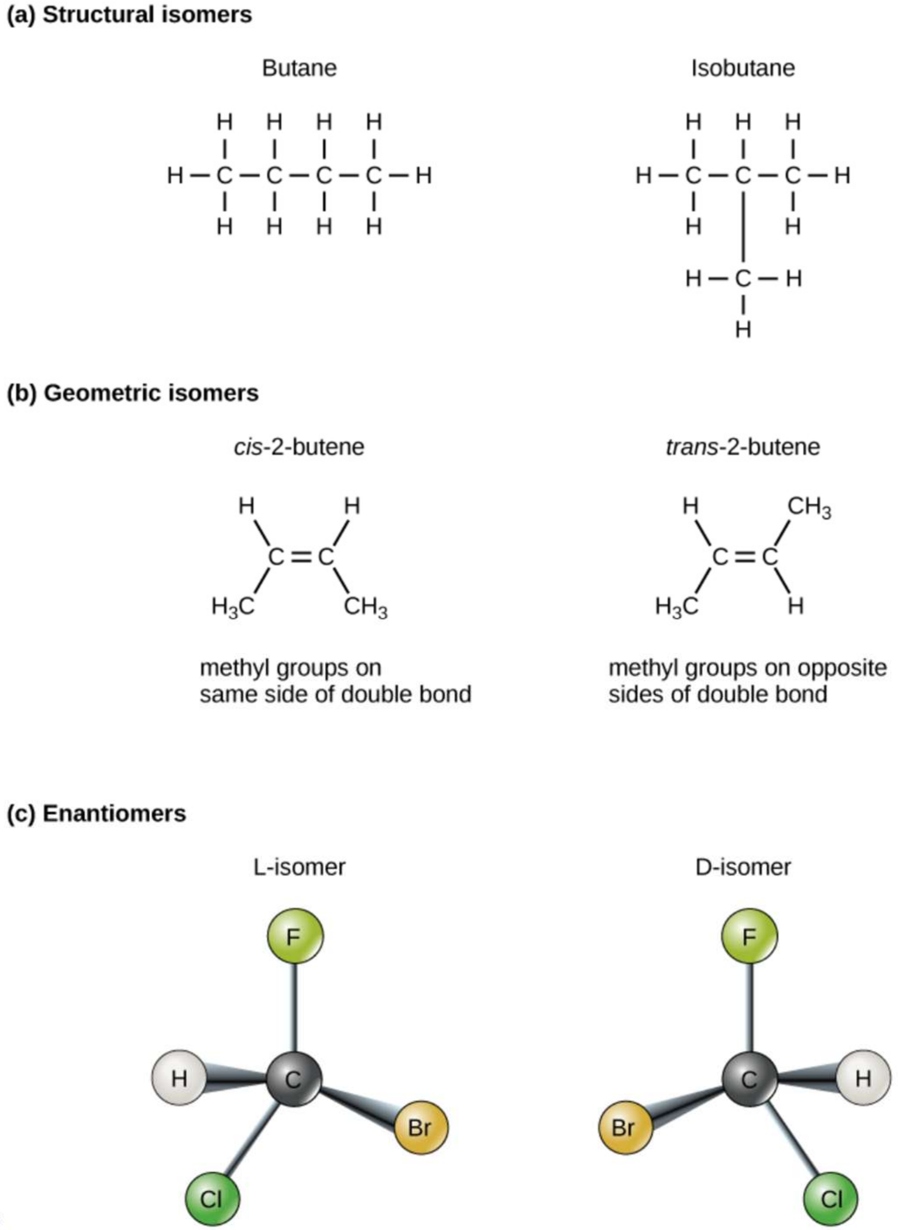


Hydrocarbons

- Functional Groups

Functional groups are groups of atoms that occur within molecules and confer specific chemical properties to those molecules.

The functional groups in a macromolecule are usually attached to the carbon backbone at one or several different places along its chain and/or ring structure.

These groups play an important role in forming molecules like DNA, proteins, carbohydrates, and lipids.

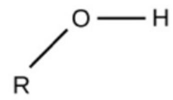
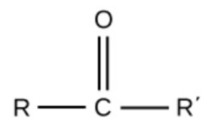
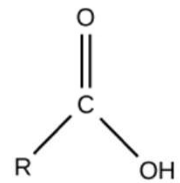
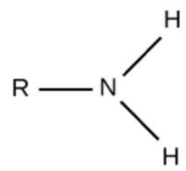
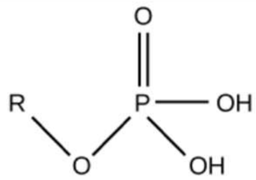
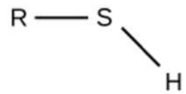
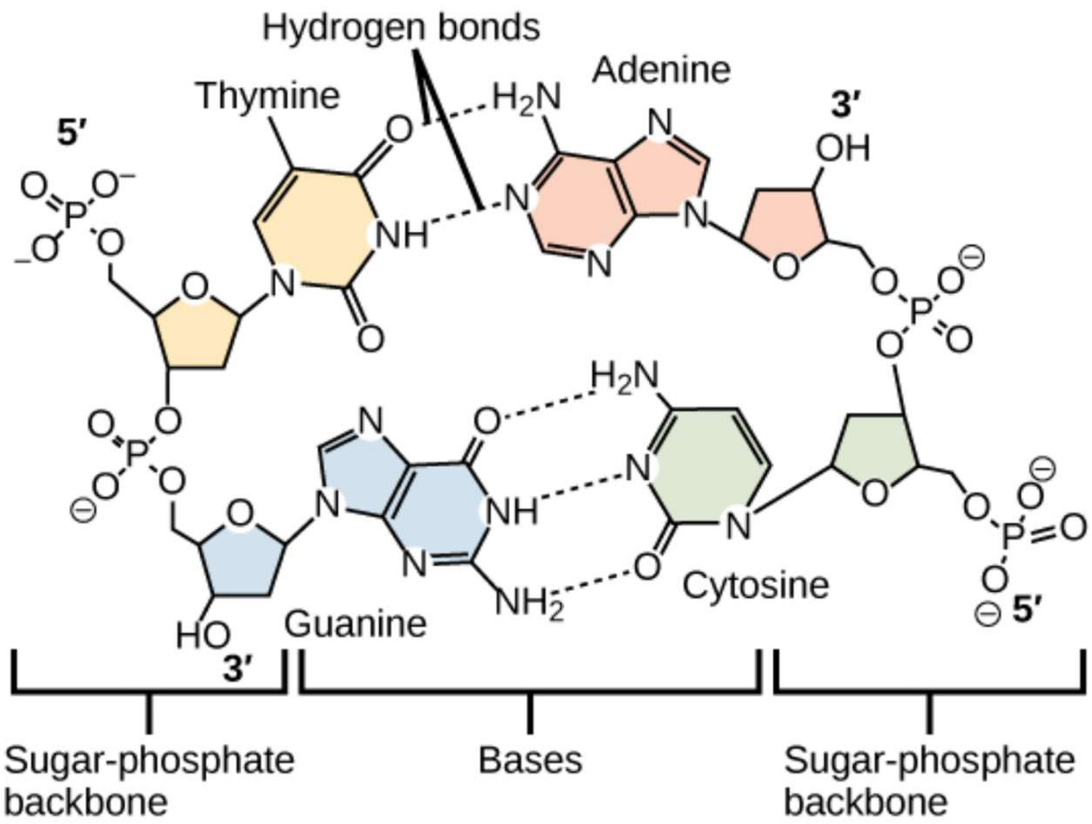
Functional Group	Structure	Properties
Hydroxyl		Polar
Methyl	R — CH_3	Nonpolar
Carbonyl		Polar
Carboxyl		Charged, ionizes to release H^+. Since carboxyl groups can release H^+ ions into solution, they are considered acidic.
Amino		Charged, accepts H^+ to form NH_3^+. Since amino groups can remove H^+ from solution, they are considered basic.
Phosphate		Charged, ionizes to release H^+. Since phosphate groups can release H^+ ions into solution, they are considered acidic.
Sulfhydryl		Polar

Figure: https://commons.wikimedia.org/wiki/File:Hot_Watering_Hole_Action.jpg

Hydrocarbons



Hydrogen bonds connect two strands of DNA together to create the double-helix structure.

Why are molecules of water so important?

- **Water** is essential to life
  - This is why the search for life on other planets often begins with a search for water
  - Some properties of water that make it able to support life:
    - Water is cohesive
    - Water has good heat capacity
    - Many substances dissolve in water
    - Water solutions come in a range of pH values

Figure: https://commons.wikimedia.org/wiki/File:Hot_Watering_Hole_Action.jpg

- [What does water do for you?]

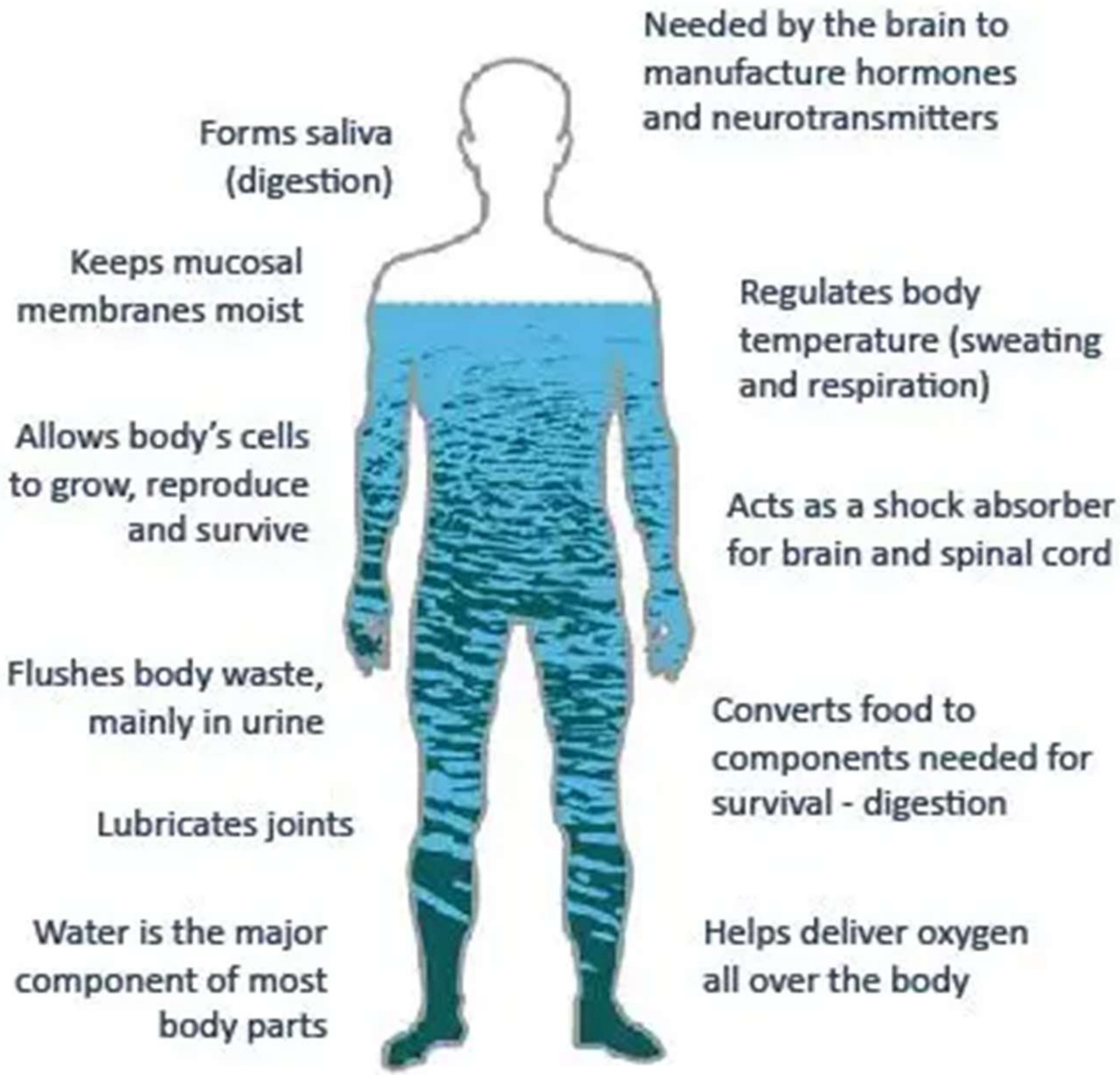
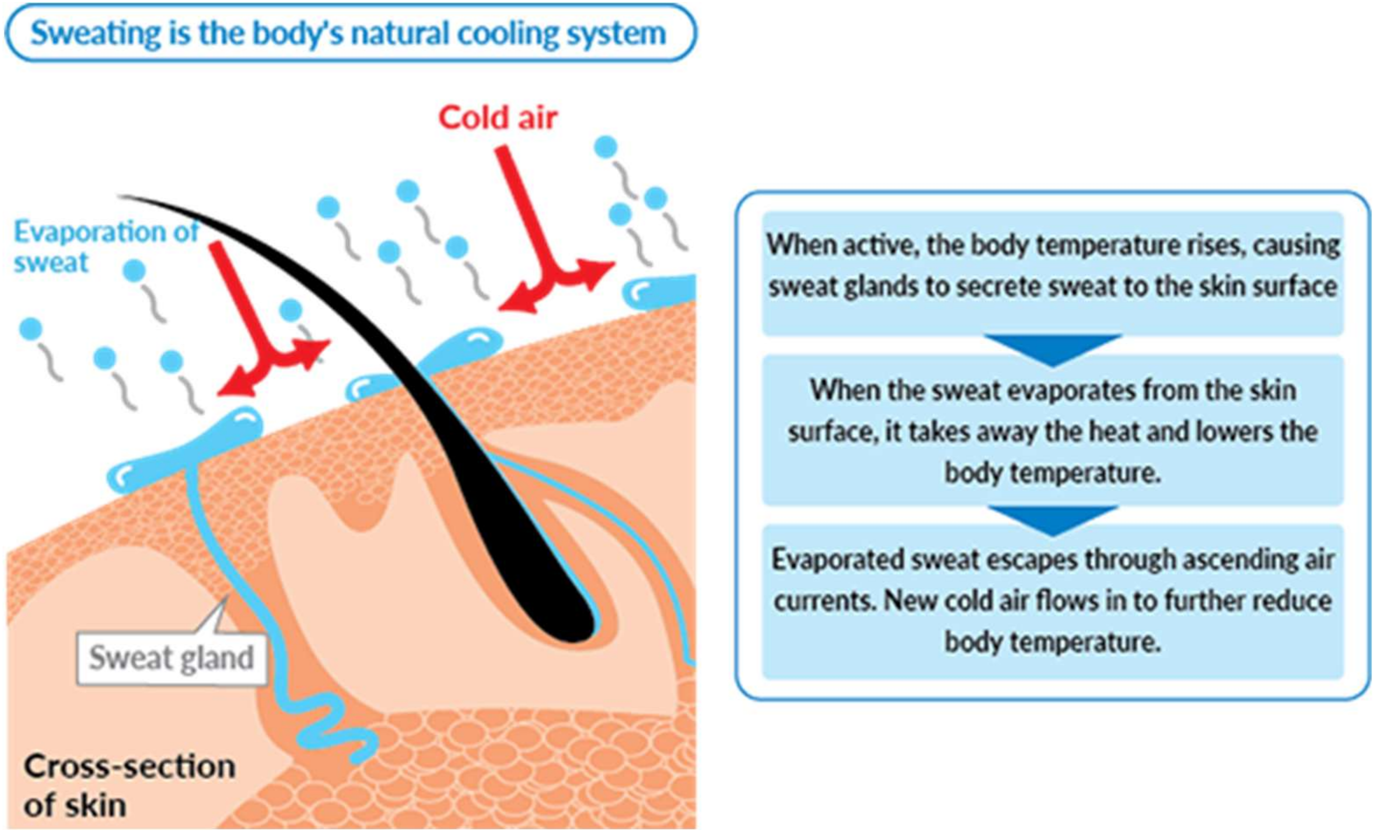


Figure: https://freesvg.org/green-tree-clip-art-vector

▪ Cooling our body



High heat capacity
High heat of vaporization 2400kJ/kg!

Figure: https://freesvg.org/green-tree-clip-art-vector

- <u>Water is cohesive</u>
  - **Cohesion** = the tendency of water molecules to stick together through hydrogen bonds
    - *remember hydrogen bonds!*
    - allows water to move to the top of the tallest trees, defying gravity

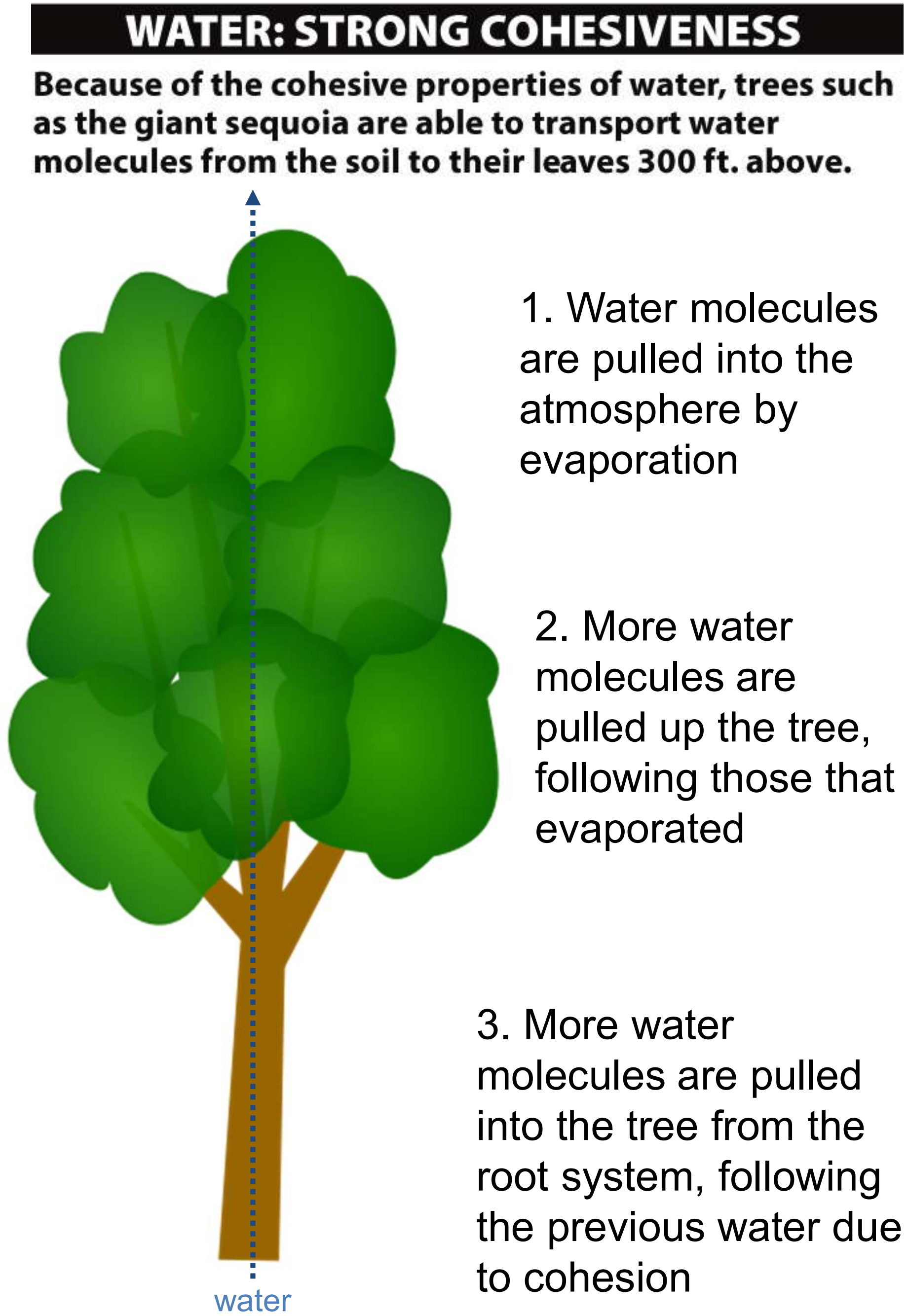
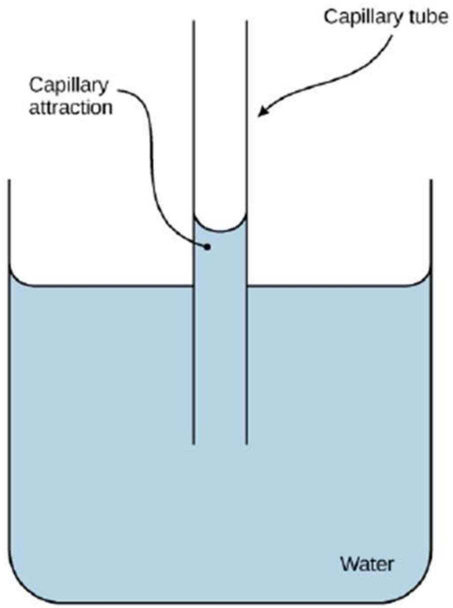


Figure: https://freesvg.org/green-tree-clip-art-vector

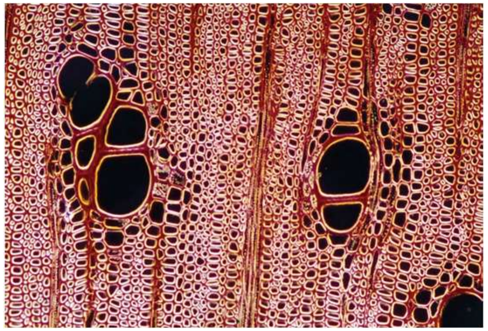
- Water cohesion also allows organisms to move across it
  - – Also why belly flops hurt!

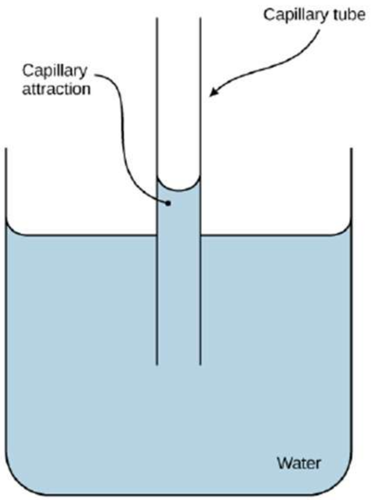
https://www.maxpixel.net/Water-Biug-Water-Wasserlaufer-Insect-Bug-5372086

https://www.flickr.com/photos/lewishamdreamer/4060481490



- How does water flow to top of tree of ~100 meters?





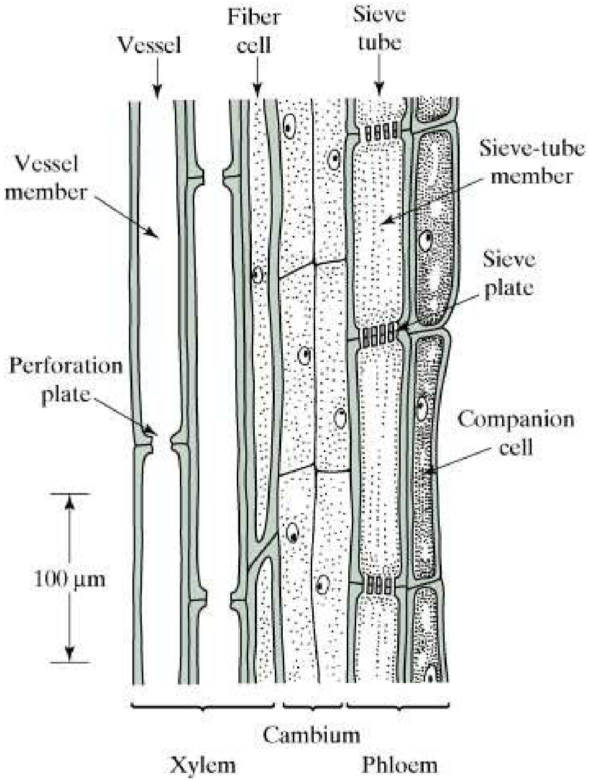


Figure from OpenStax Biology 2e

- <u>Many substances dissolve in water</u>
  - Substances that dissolve in water = **hydrophilic**
    - "water-loving"
    - e.g. salts, sugars, & electrolytes

      – important because we need these things to move through our blood stream (which is mostly water) easily

Notice how water molecules happily surround these salts, allowing them to move in the blood stream easily

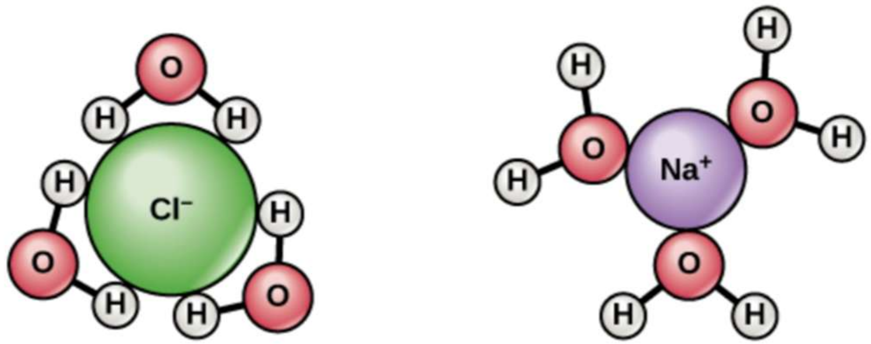
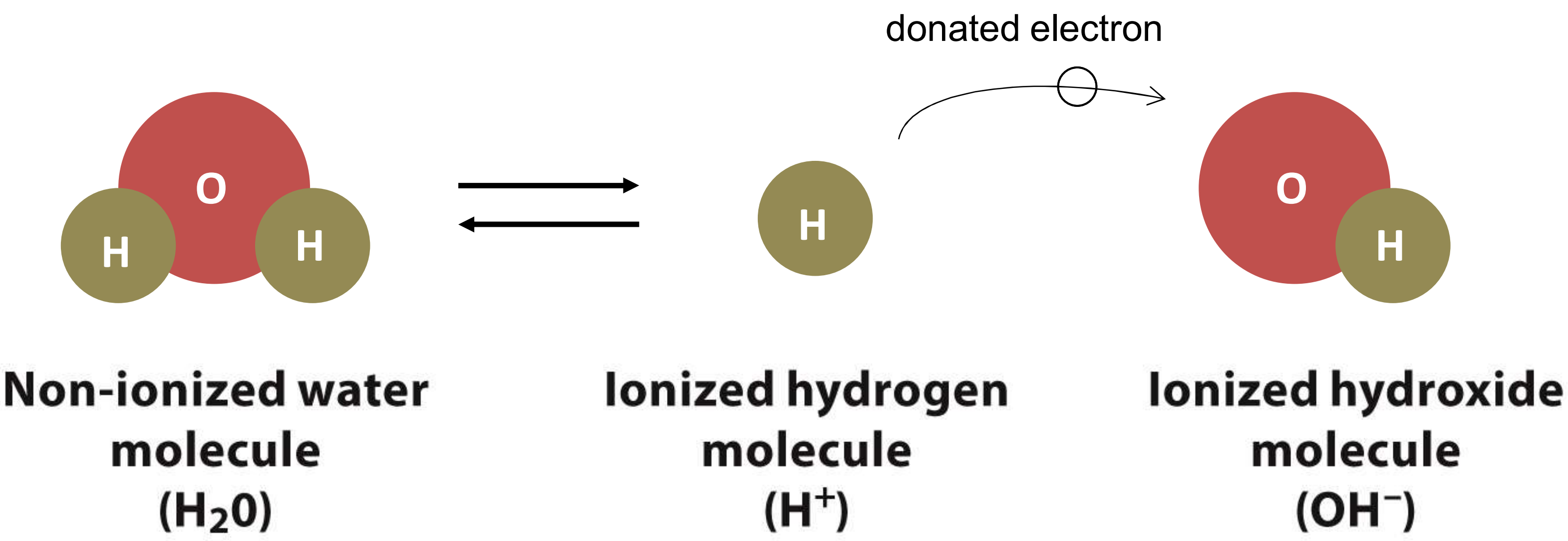


Figure from OpenStax Biology 2e

- <u>Water solutions come in a range of pH values</u>
  - In a solution of water, some molecules become "ionized" – they break apart & become charged
    - $H_2O \rightarrow OH^- + H^+$



cation *anion*

- Solutions where $OH^- = H^+$ are **neutral** *(equal)*
  - pH 7
  - e.g. pure water

- Solutions where $OH^- < H^+$ are **acidic** *(more H+)*
  - pH 0-6
  - e.g. lemon juice & vinegar

- Solutions where $OH^- > H^+$ are **basic** *(more OH-)*
  - pH 8-14
  - e.g. ammonia & bleach

- The **pH scale** is a way of referring to the acidic, basic, or neutral quality of a solution

BASES

Bases are fluids that have a greater proportion of OH^- ions to H^+ ions.

- OH^- ions bind with H^+ ions, neutralizing acids.
- Strong bases are caustic to your skin.
- Bases can be found in many household cleaners.
- Bases are generally bitter in taste and slippery.

ACIDS

Acids are fluids that have a greater propotion of H^+ ions to OH^- ions.

- H^+ ions are very reactive.
- Strong acids are corrosive to metals.
- Acids break down food in your digestive tract.
- Acids are generally sour in taste.

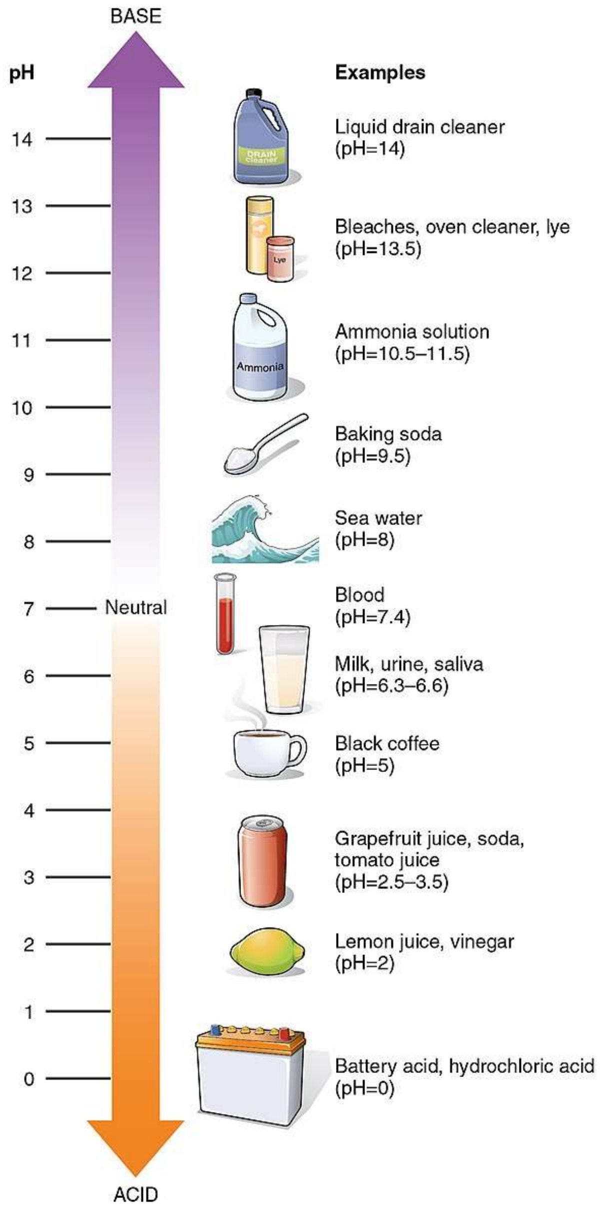


Figure: https://commons.wikimedia.org/wiki/File:216_pH_Scale-01.jpg

- Many substances dissolve in water – some don't
  - Substances that do NOT dissolve in water = **hydrophobic**
    - "water-fearing" – clump up in water
    - e.g. fats, oils, & cell membranes
      - it is important that our cells do not dissolve when we get wet, or when they contact our watery blood stream, so we use these things to waterproof ourselves & form barriers

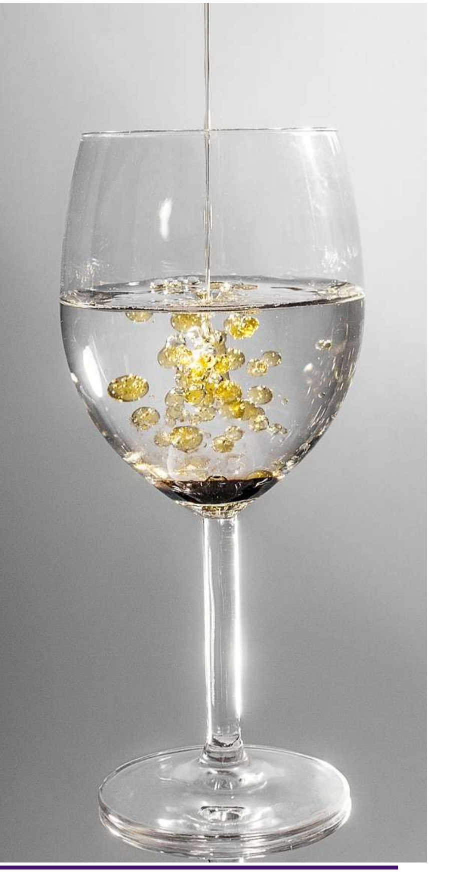


Figure: https://www.maxpixel.net/Water-Drink-Crystal-Glass-Oil-Liquid-Drip-Glass-101666

Hydrophobic Force

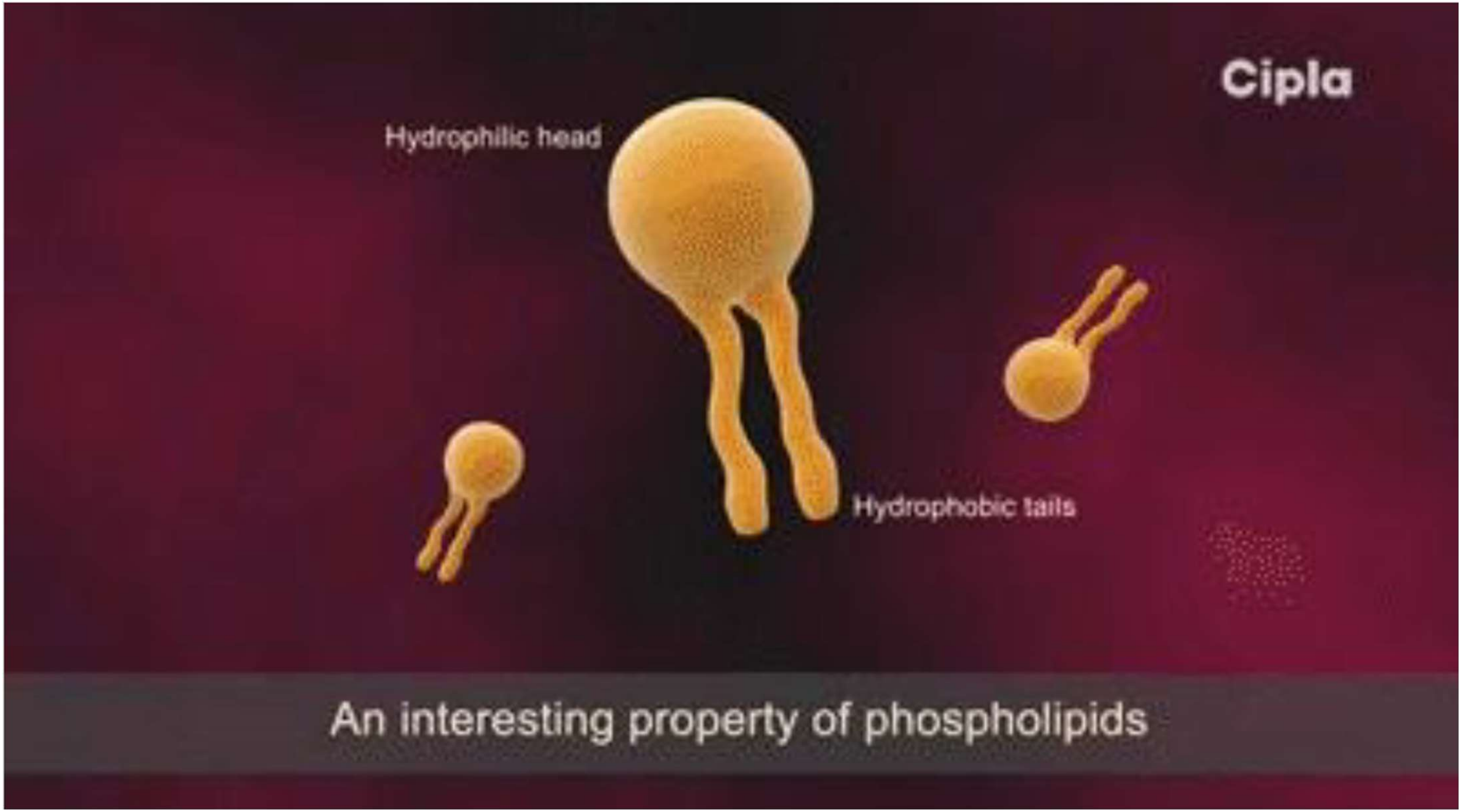
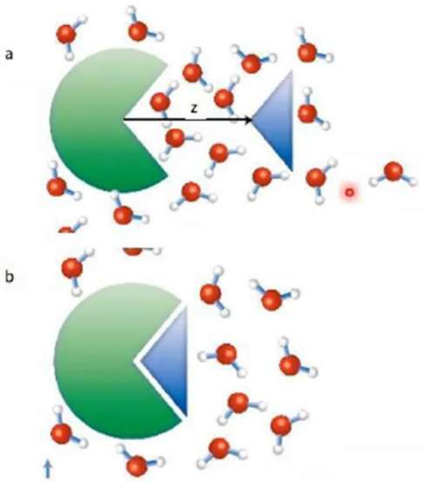
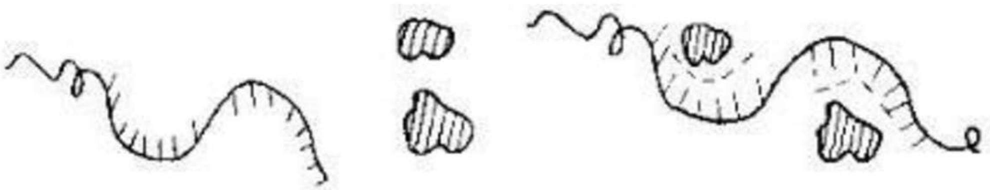
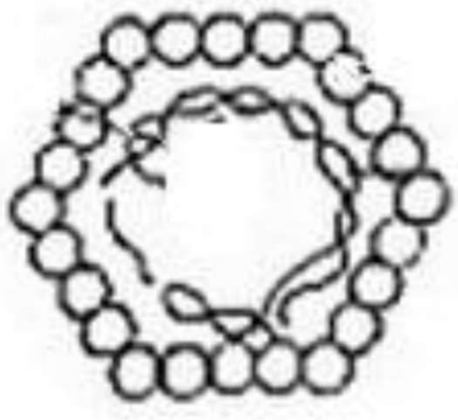


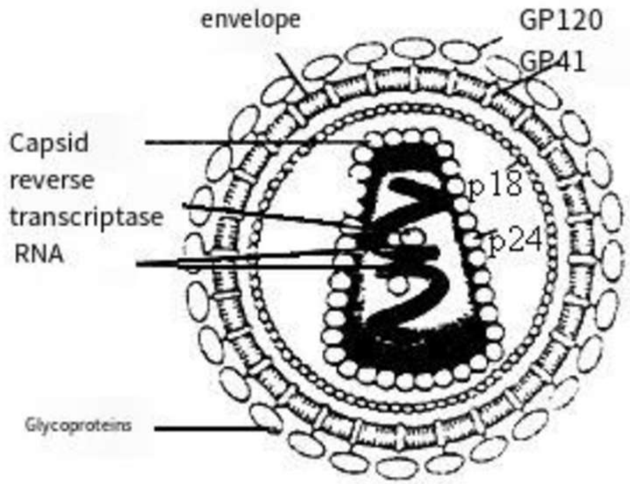
Figure: https://www.maxpixel.net/Water-Drink-Crystal-Glass-Oil-Liquid-Drip-Glass-101666

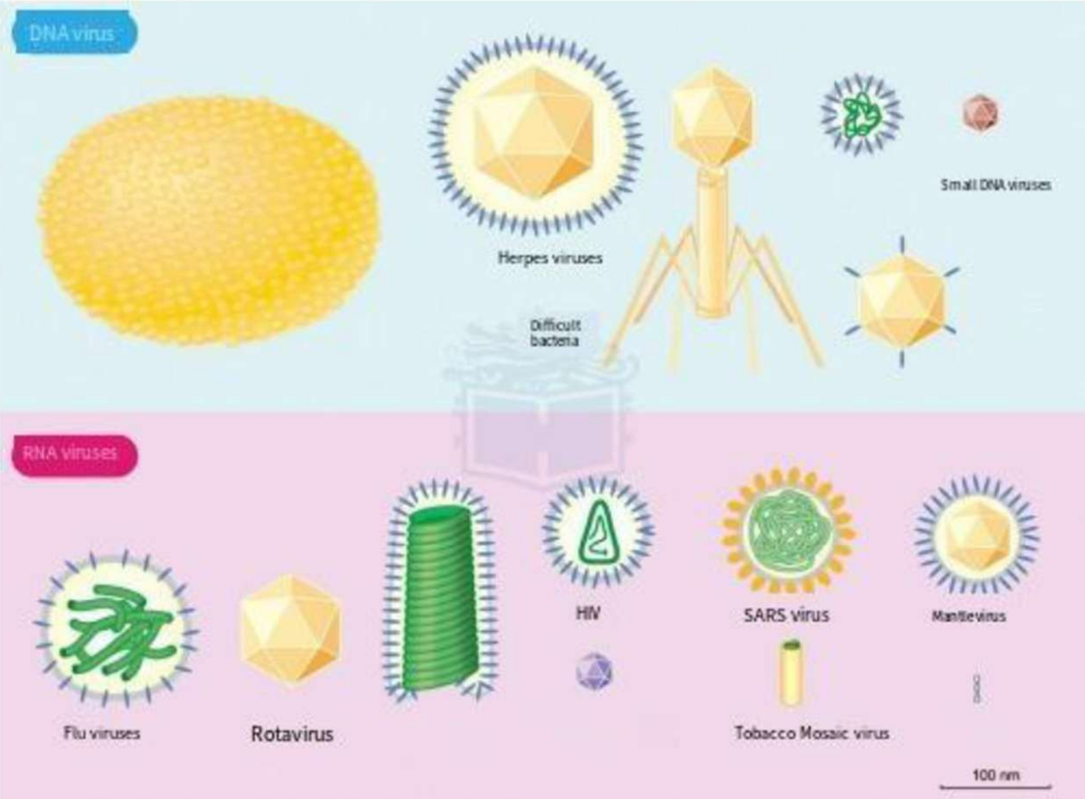


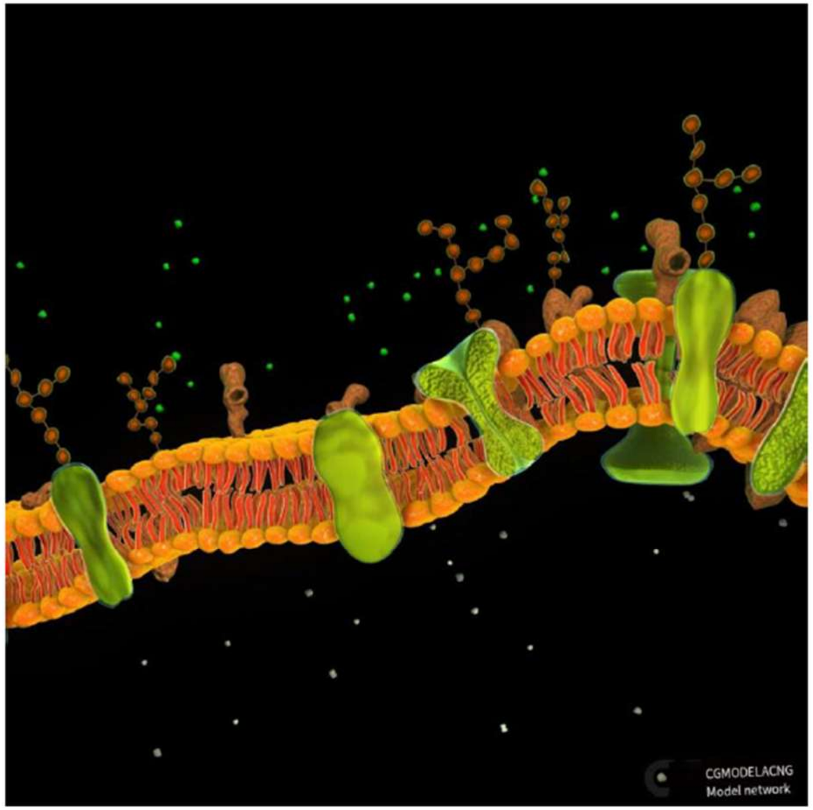


Hydrophobic action is when spherical proteins fold up in water and hide their hydrophobic parts inside. You could also say that this process changes the free energy because it increases entropy by releasing bound water during interactions between molecules.






DNA virus
Herpes viruses
Small DNA viruses
Difficult bacteria
RNA viruses
Flu viruses
Rotavirus
HIV
SARS virus
Mantlevirus
Tobacco Mosaic virus
100 nm


CGMODELACNG
Model network

- Water will come up over & over as we learn more about how our cells & bodies work
  - We'll talk about the other molecules that make up our bodies – how they interact with water will be important (e.g. hydrophilic vs. hydrophobic)
  - We'll talk about why it's so important to keep our blood neutral, & not acidic or basic

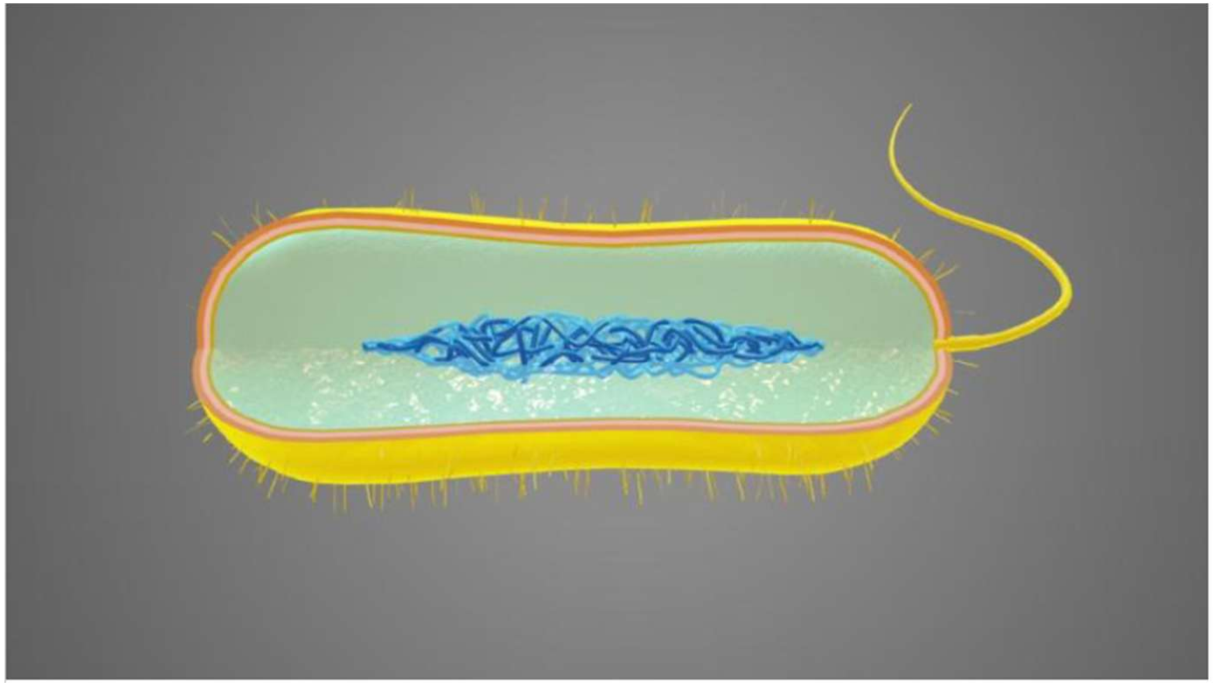
- Up next: water is essential to life & our bodies contain a lot of it, but it is not what living things are actually built of
  - **So what molecules are we built of?**

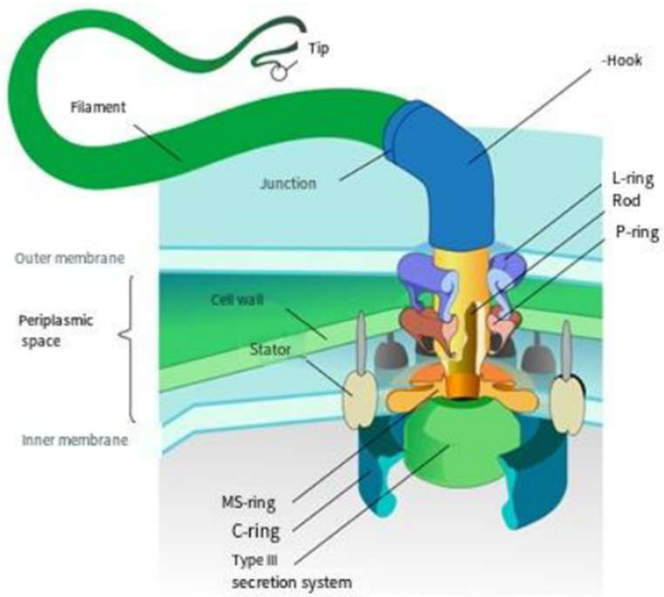
Chapter Summary

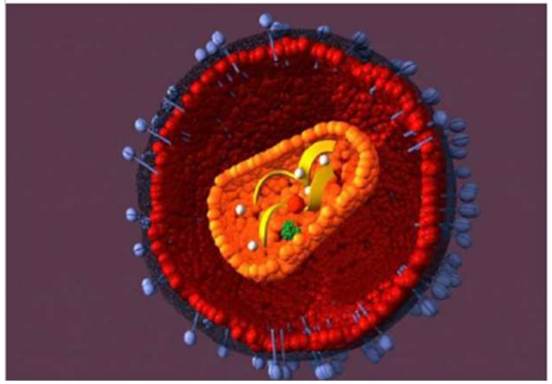
- The building blocks：　atoms, isotopes, ions, molecules
- Water
- Carbon and Hydrocarbons

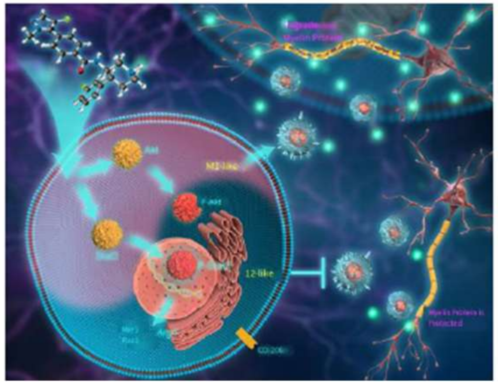
Figure: https://commons.wikimedia.org/wiki/File:Hot_Watering_Hole_Action.jpg

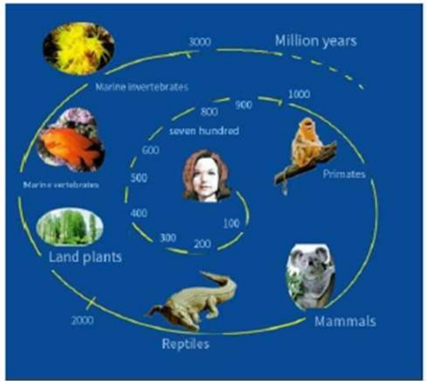

The origin and evolution of the universe
13.5 billion years ago, the Big Bang created matter, energy, time and space in the universe. (Physics)
After 300,000 years, matter and energy form complex structures and atoms and molecules appear (chemistry)
3.8 billion years ago, molecules joined together to form organisms (biology)

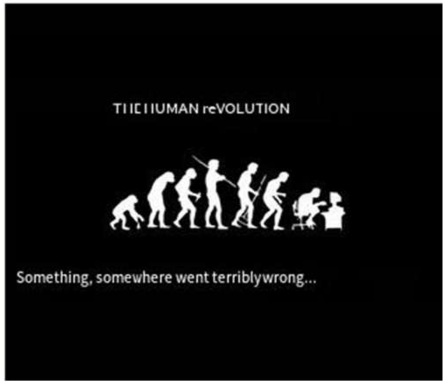



Tip
Hook
Filament
Junction
L-ring
Rod
P-ring
Outer membrane
Periplasmic space
Cell wall
Stator
Inner membrane
MS-ring
C-ring
Type III secretion system






3000
Million years
Marine invertebrates
1000
900
800
seven hundred
600
500
400
300
200
100
Primates
Marine vertebrates
Land plants
2000
Reptiles
Mammals


THE HUMAN reVOLUTION
Something, somewhere went terriblywrong...